3. Aircraft Life Cycle and Development



3. Aircraft Life Cycle and Development

3.1 Consumer Needs

3.2 Aircraft Life Cycle

3.3 Aircraft Design Process

3.4 Aircraft Cabin Design

3.5 Costs

3. Aircraft Life Cycle and Development

3.1 Consumer Needs



Key Buying Factors

Leisure Traveler

Business Traveler



Source: Conrady (2013)

General Types of Passengers



- Highly price-sensitive
- Comfort is less important
- Check-in time max. 3 hours before departure is accepted
- Flight is part of the vacation experience
- Lots of baggage
- Experience of in-flight entertainment



- Flexibility more important than ticket price
- Requires quick check-in
- Stand-by periods should be kept as short as possible
- Carries a lot of hand baggage
- Requires in-flight communication as travel time = work time
- Values comfort and service

The Six Traveller Tribes of the Future (1)

2020



Active Seniors
50-75 years in 2020
(born 1945-1970)



Global Clans



Cosmopolitan Commuters



Global Executives











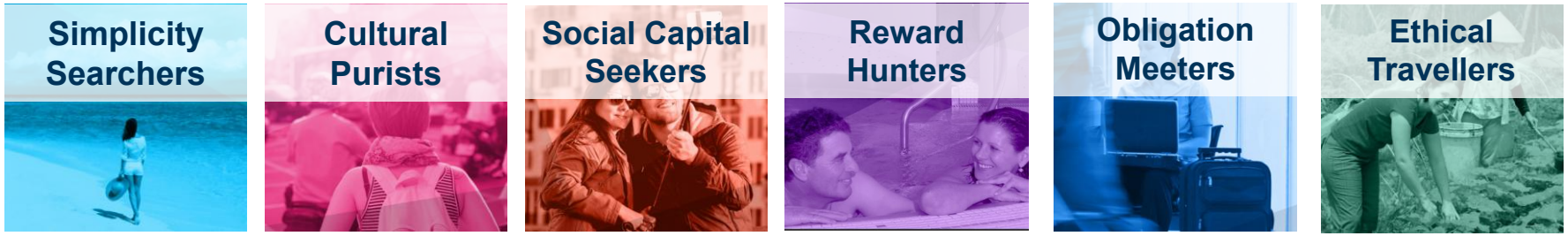


2030

Source: Amadeus/Future Traveller Tribes 2030

The Six Traveller Tribes of the Future (1)

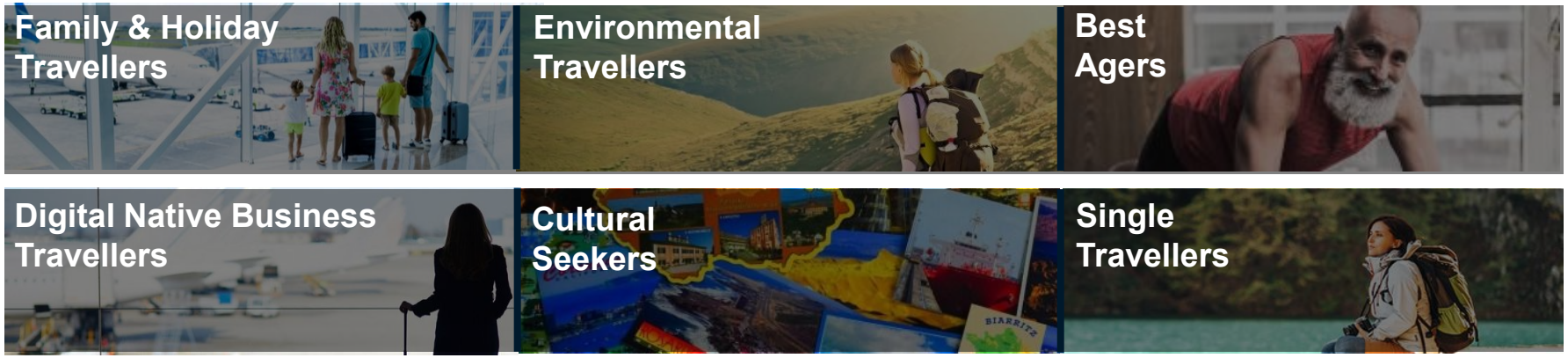
Amadeus Future Traveller Tribes 2030



Different Studies

… Same Travellers

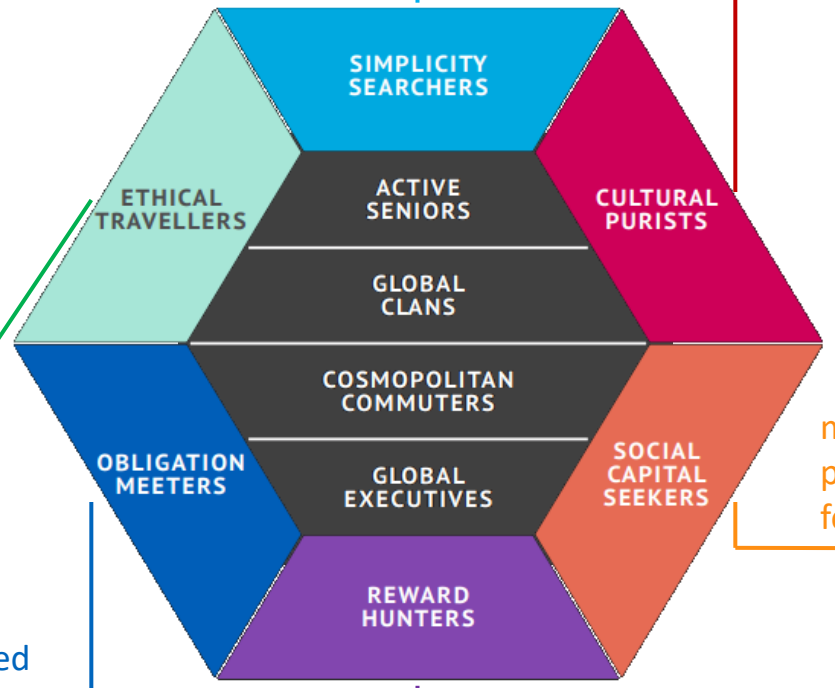
Kluge et al.: Profiling Future Air Transport Passengers in Europe



The Six Traveler Tribes of the Future (2)

ease, transparency, outsourcing decision-making to trusted parties

immerse themselves in an unfamiliar culture, engage sincerely with a different way of living



environmentally concerned, shaped by political ideals, high awareness of their impact on the visited economy

maximum social reward, exploit potential of digital media, visibility for online audiences

mostly business travellers, bounded objective, need to be in a certain place, at a certain time, without fail

busy lives, seek return on the investment, extraordinary and indulgent experiences

Source: Amadeus/Future Traveller Tribes 2030

Consumer Trends (1)

- **Green responsibilities**. Regardless of how climate science evolves, **eco-ethical concerns will be a feature of the future consumer landscape**. For many they will inspire micro-concessions to **less wasteful or luxurious (travel) behaviours**.

- **Work-life compounds**. Flexi-time, zero-hours contracts, increased opportunities for self-employment, and a **broad casualisation of business practices and arrangements** have given more choice about when we travel for work. This is producing a hybrid form of travel – often referred to as "**bleasure**" – for which work takes on some of the features of play, and play some of the features of work.

- **Option shock**. The **unfathomable amount of information** in the digital space is driving a market for engines and agents to condense and package choices into "**bundles**", which make comparing options easier.

- **Enterprise networking**. As social reaches deeper into the lives of billions of people, we will see **online networking more thoroughly assert its professional importance**. Consumers will invest more time and energy in **building for themselves lucrative online brands**, and reap new kinds of commercial returns on this investment.

Source: Amadeus/Future Traveller Tribes 2030

Consumer Trends (2)

- **Peer power**. The **breadth of expertise available from the networks** of friends, family members and fellow consumer-citizens online have produced a revolution in our **preferred advisory sources**, and are **fuelling an emergent peer economy**.

- **Narrative data**. Big Data will allow brands to tell **ever more focused and compelling stories** about our lives, **personalising services** not only efficiently, but imaginatively.

- **Pricing gets personal**. Tariff-setting will not be excluded from the Big Data domain. **Perfect Price Discrimination** will allow travel brands to open up their offerings to **new, less affluent emergent tourist groups**.

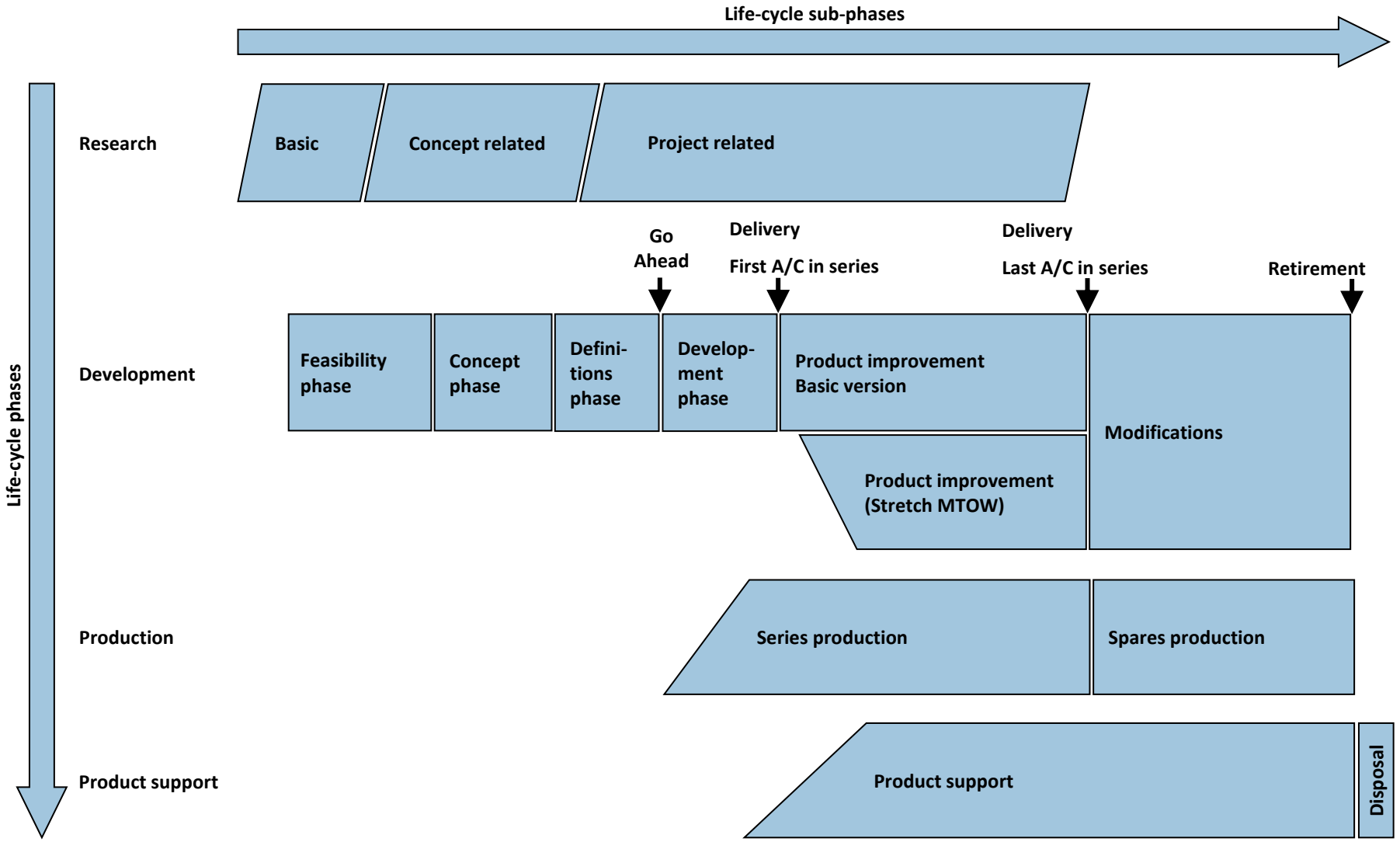
Source: Amadeus/Future Traveller Tribes 2030

3. Aircraft Life Cycle and Development

3.2 Aircraft Life Cycle



Aircraft Life Cycle: Overview



Source: Schmitt (1999)

Operating Life Cycle

Example: Airbus A300B4-622R (MSN555)



- 12 January 1990: first flight
- 22 March 1990: delivered to ILFC & leased by Air Liberté
- 03 October 1996: returned to ILFC
- 01 December 1997: leased and operated by China Airlines
- 01 January 1998: sold to Ansett Worldwide Aviation Services, still operated by China Airlines
- 07 June 2004: sold to AWAS Aviation Capital Ltd., still operated by China Airlines
- 20 February 2006: returned to AWAS Aviation Capital Ltd.
- 13 April 2006: leased and operated by Eagle Aviation France
- 31 October 2006: leased and operated by Saudi Arabian Airlines
- 06 November 2008: purchased by FedEx, subsequent conversion to freighter
- April 2009: operated by FedEx as air freighter
- 10 April 2014: withdrawn from use, stored at Victorville Airport (California)

Source: ACAS

Operating Life Cycle

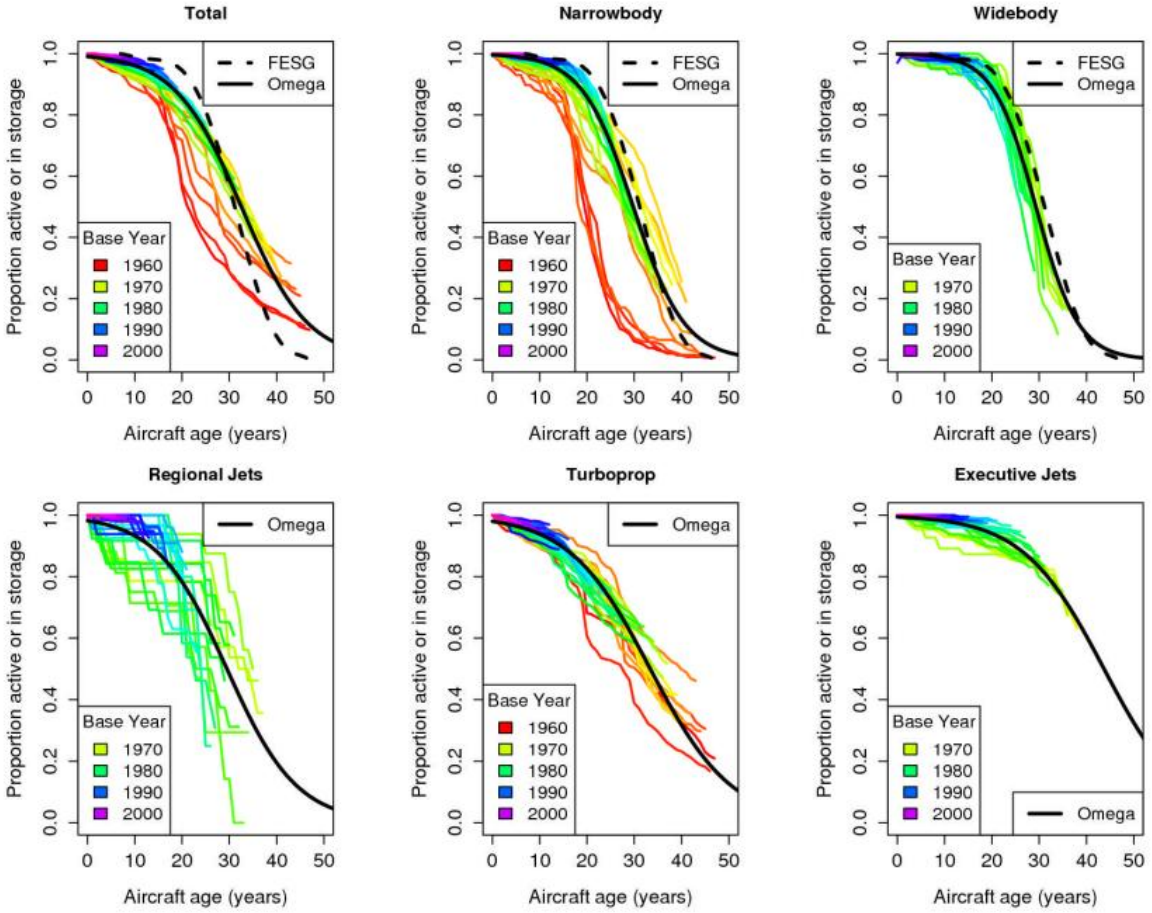
Example: Airbus A300B4-622R (MSN555)



Source: Google Maps

Retirement Curves

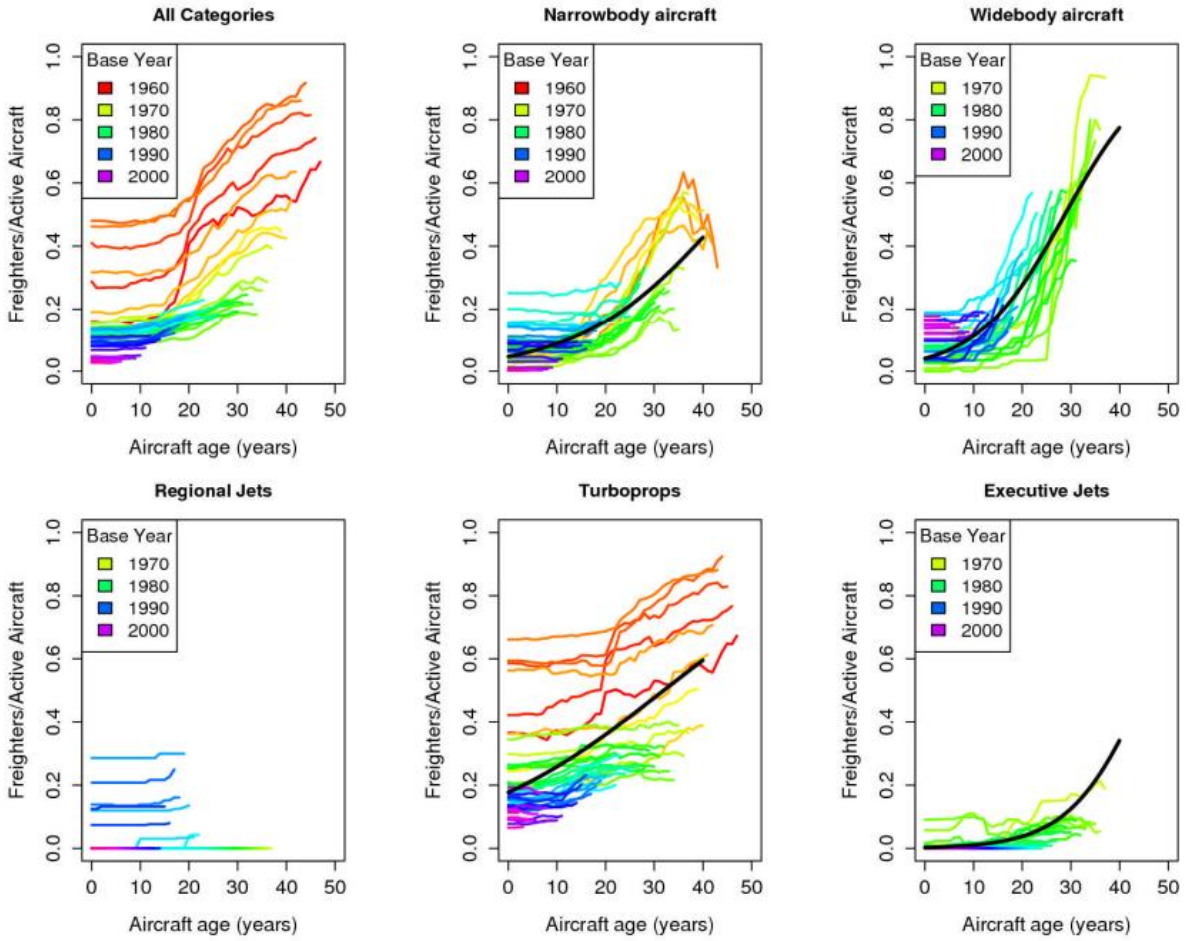
Proportion of active aircraft



Source: Morrell & Dray (2009)

Freighter Conversions

Proportion of active freighter aircraft



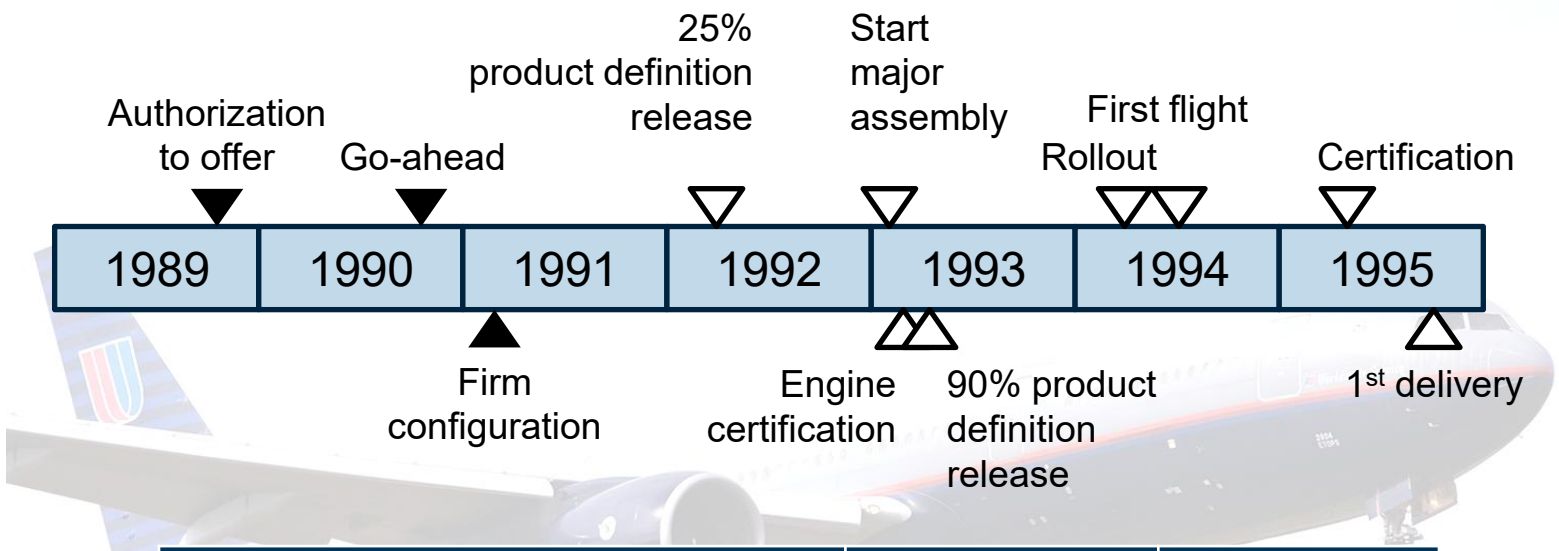
Source: Morrell & Dray (2009)

3. Aircraft Life Cycle and Development

3.2 Aircraft Design Process



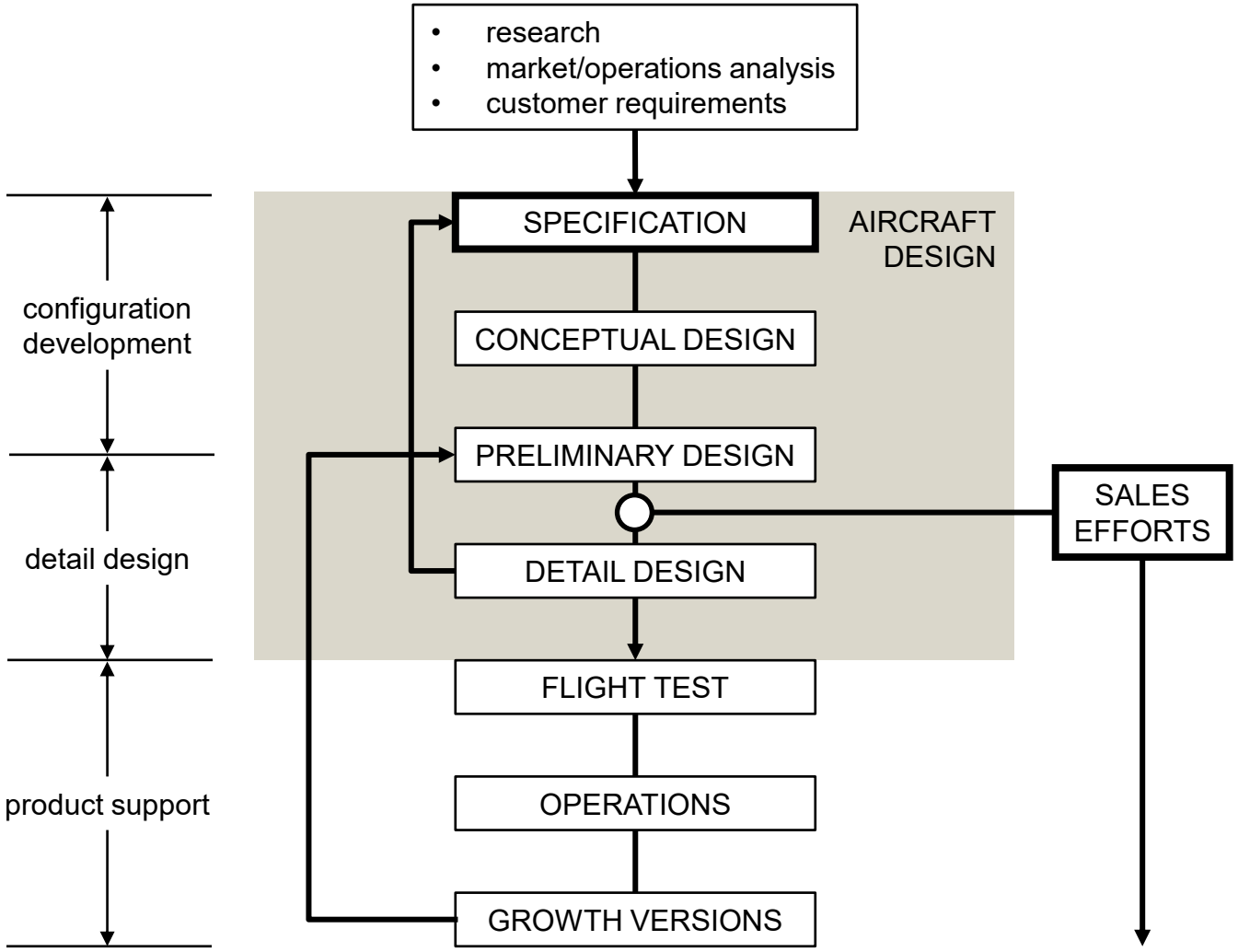
Example: Development Milestones of the Boeing 777



Event	Date	Elapsed months
Program launch (UAL signed for 34 aircraft)	Oct 1990	--
Rollout	Apr 1994	40
Full-scale static tests begin	Jun 1994	43
First flight	Jun 1994	43
Wing completes ultimate loading (destroyed)	Jan 1995	51
FAA / JAA aircraft certification	May 1995	55
First delivery (UAL)	May 1995	55
Fatigue tests complete two lifetimes	Mar 1997	83

Source: Petersen and Sutcliffe (2012), Nicolai (2005)

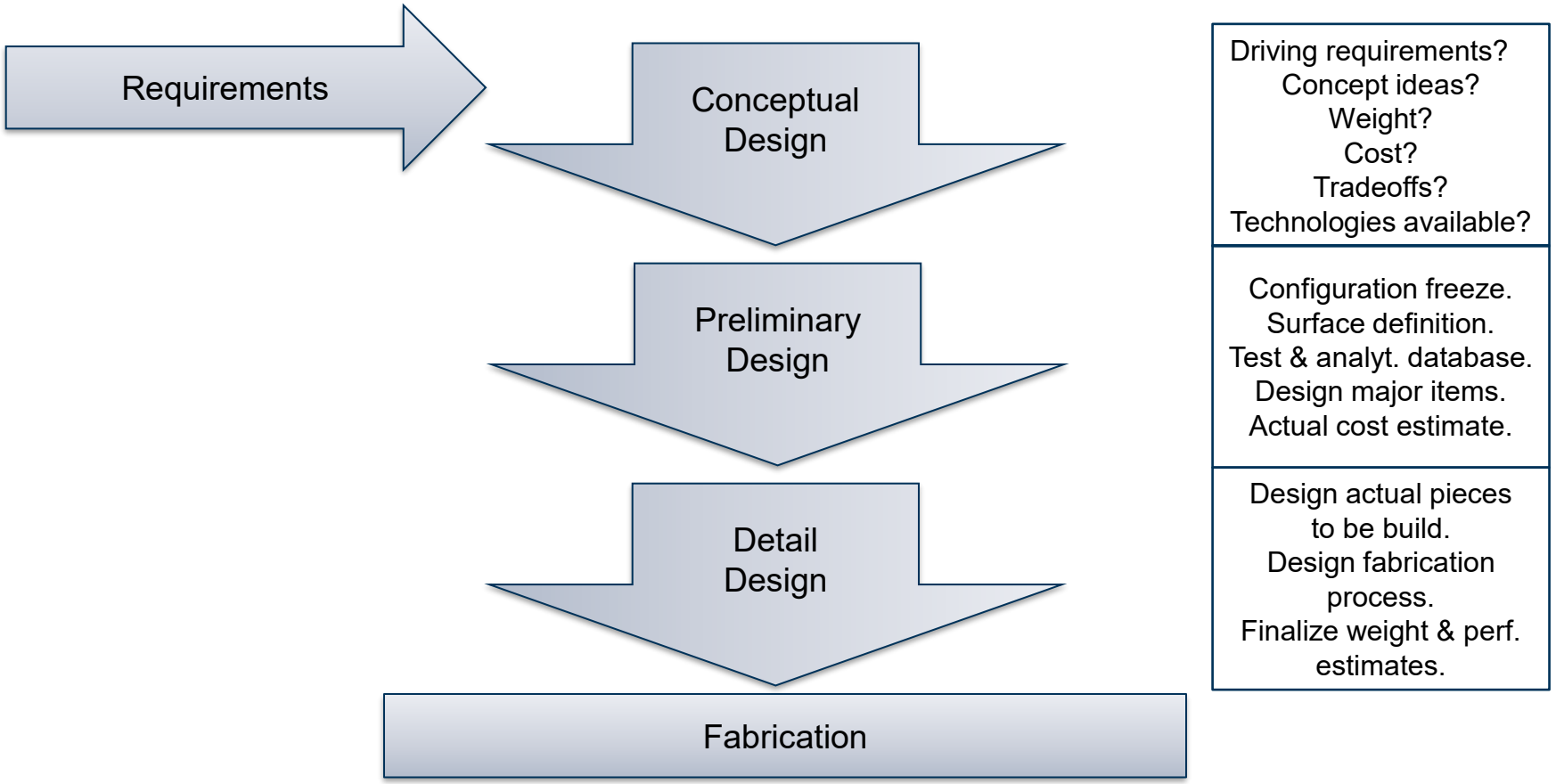
Schedule of the Civil Airplane Development Process



Source: Torenbeek (2013), McMasters (2004)

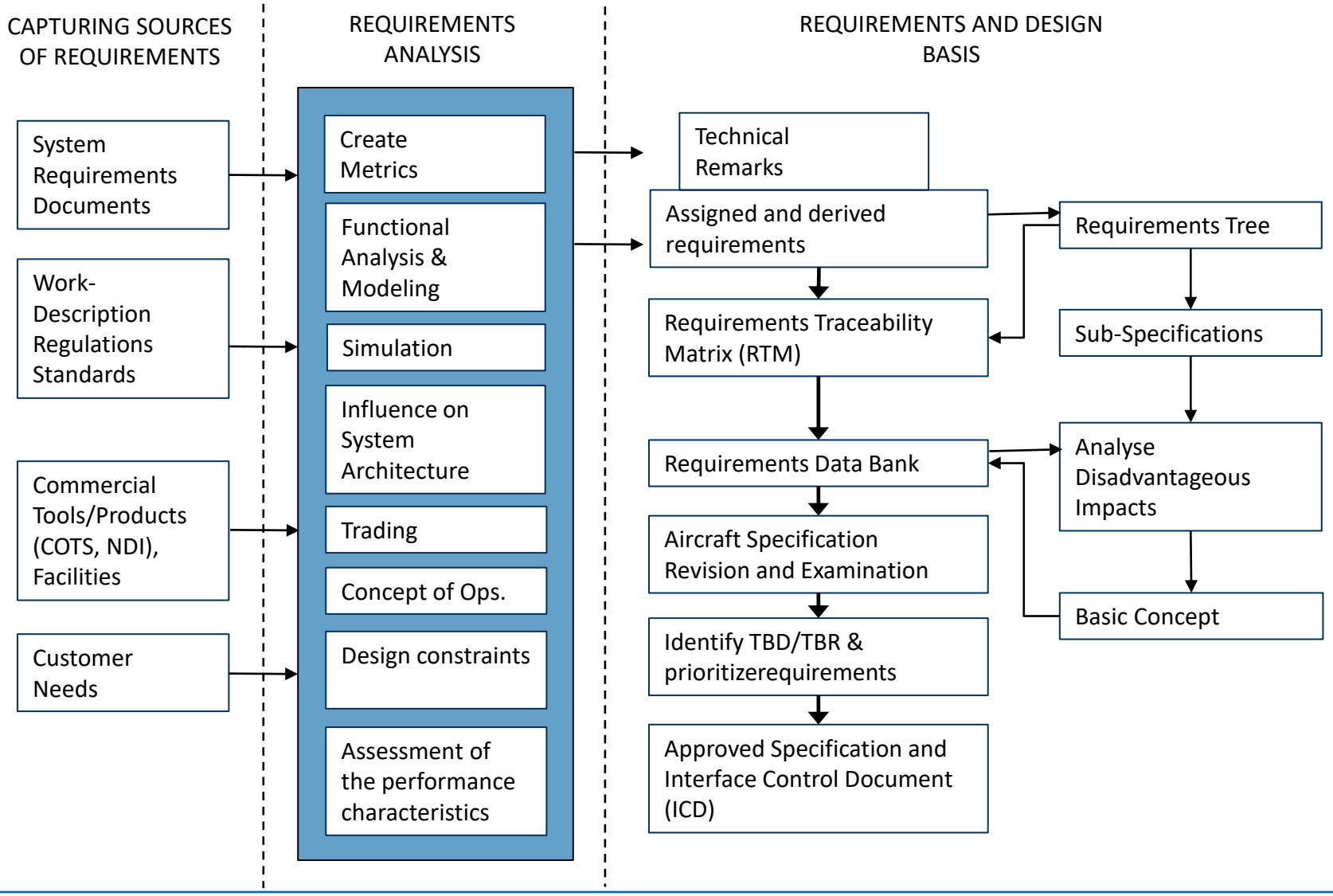
Aircraft Design: Overview

The aircraft design process is fundamentally composed of three phases.

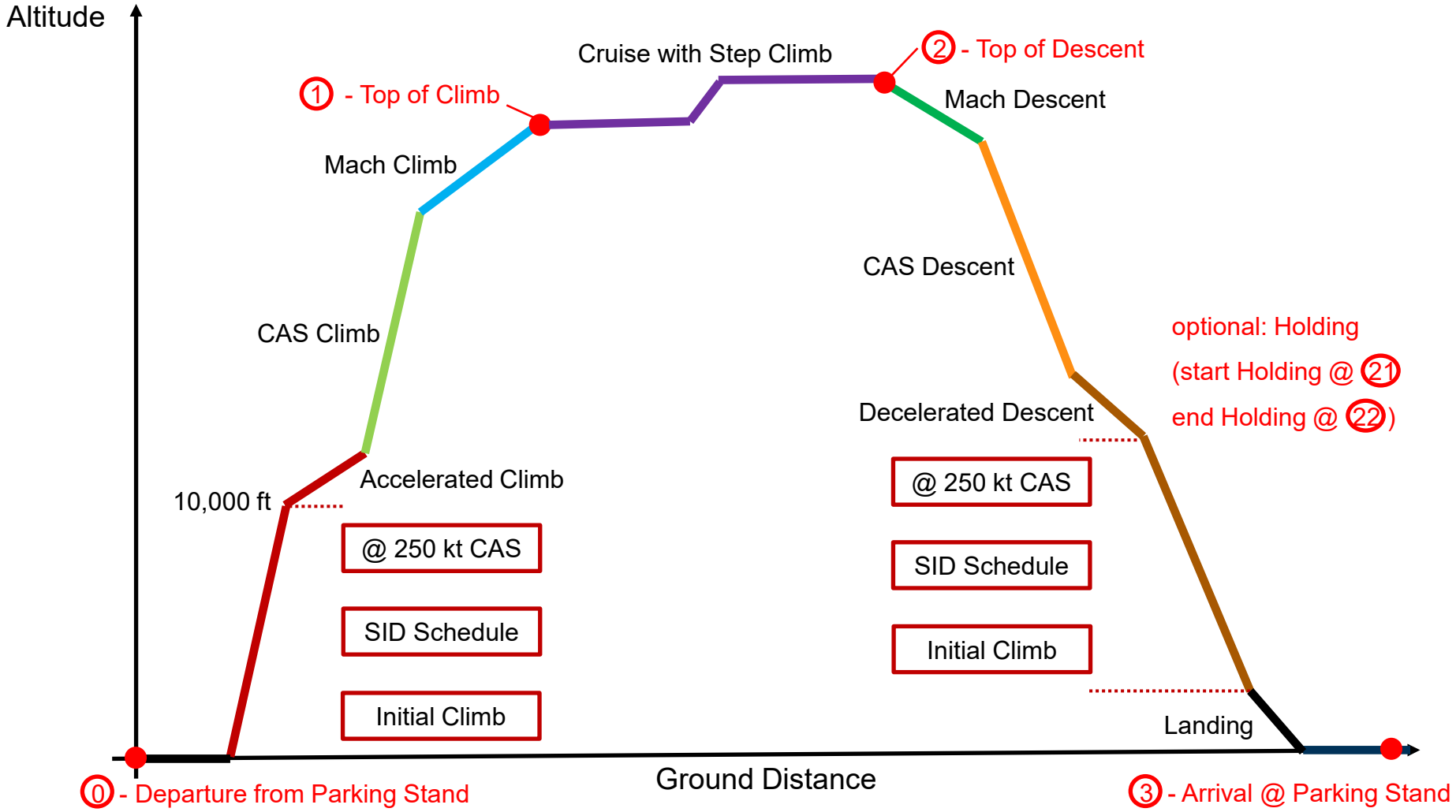


Source: Raymer 2012

Requirements Engineering (1)



Mission Analysis: Flight Phases (0)

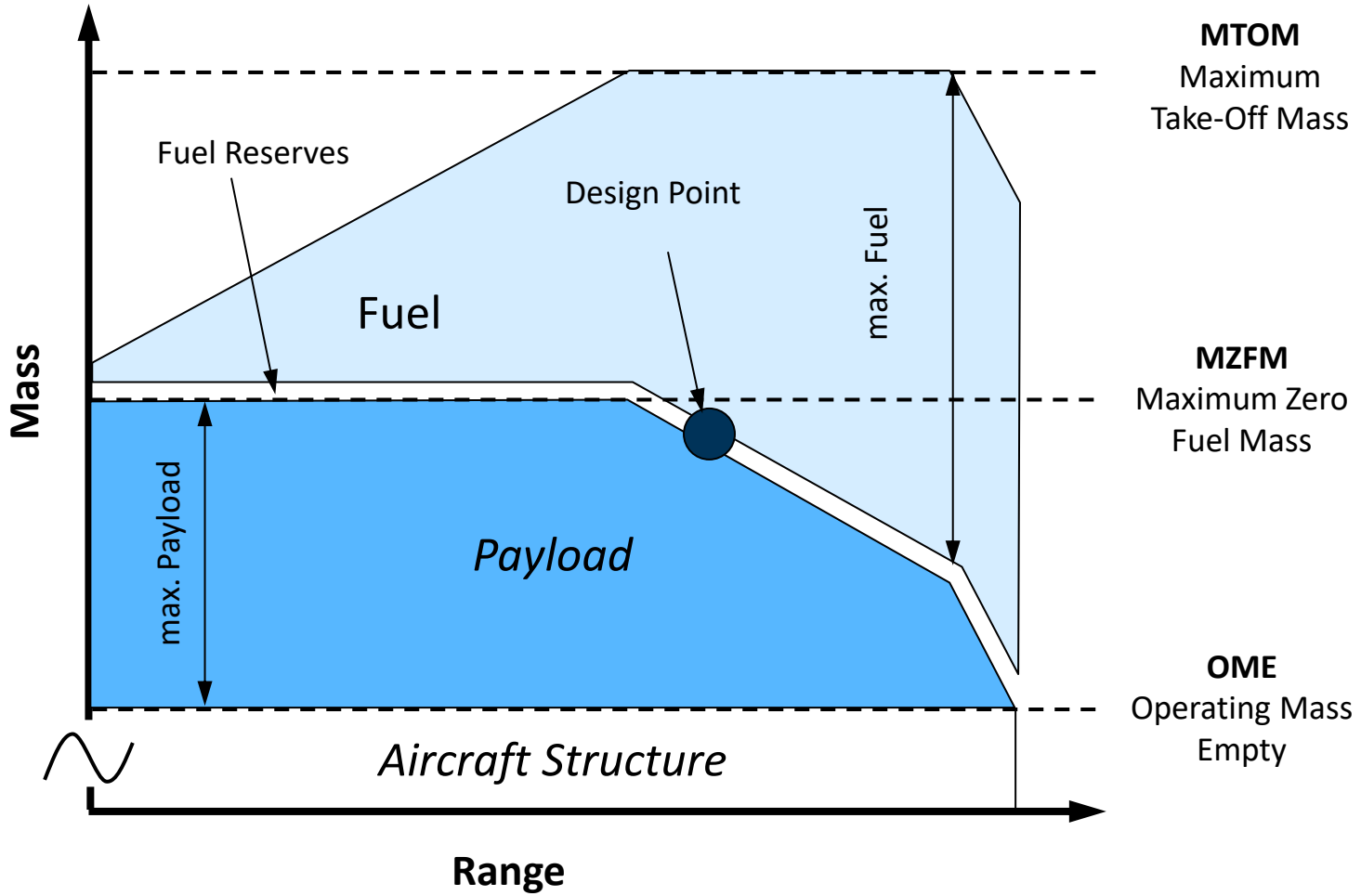


After Uzun et al.: Design of a Hybrid Digital-Twin Flight Performance Model Through Machine Learning

Mission Analysis: The Payload-Range Diagram (1)

Two major criteria: **Payload Capacity** and **Range Capability**

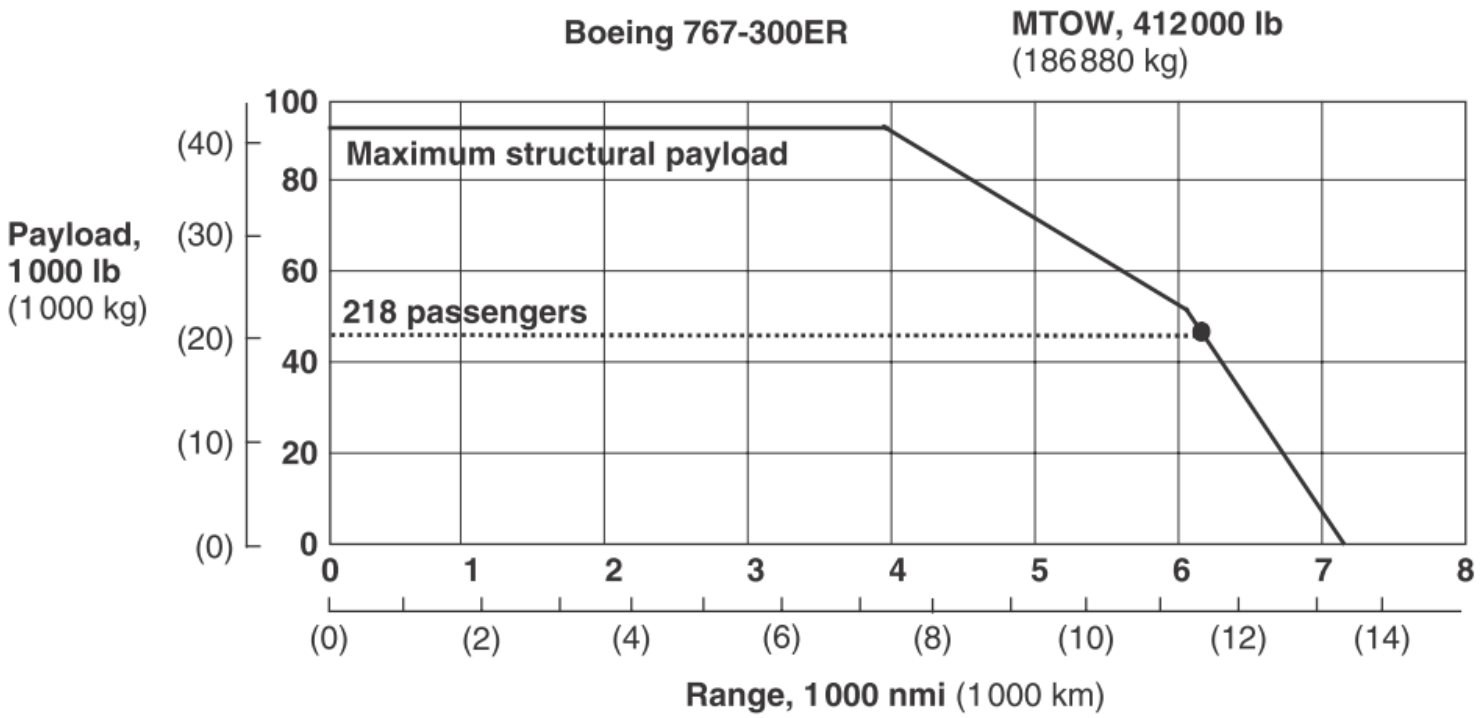
Payload-Range Diagram



Source: Boeing

Mission Analysis: The Payload-Range Diagram (2)

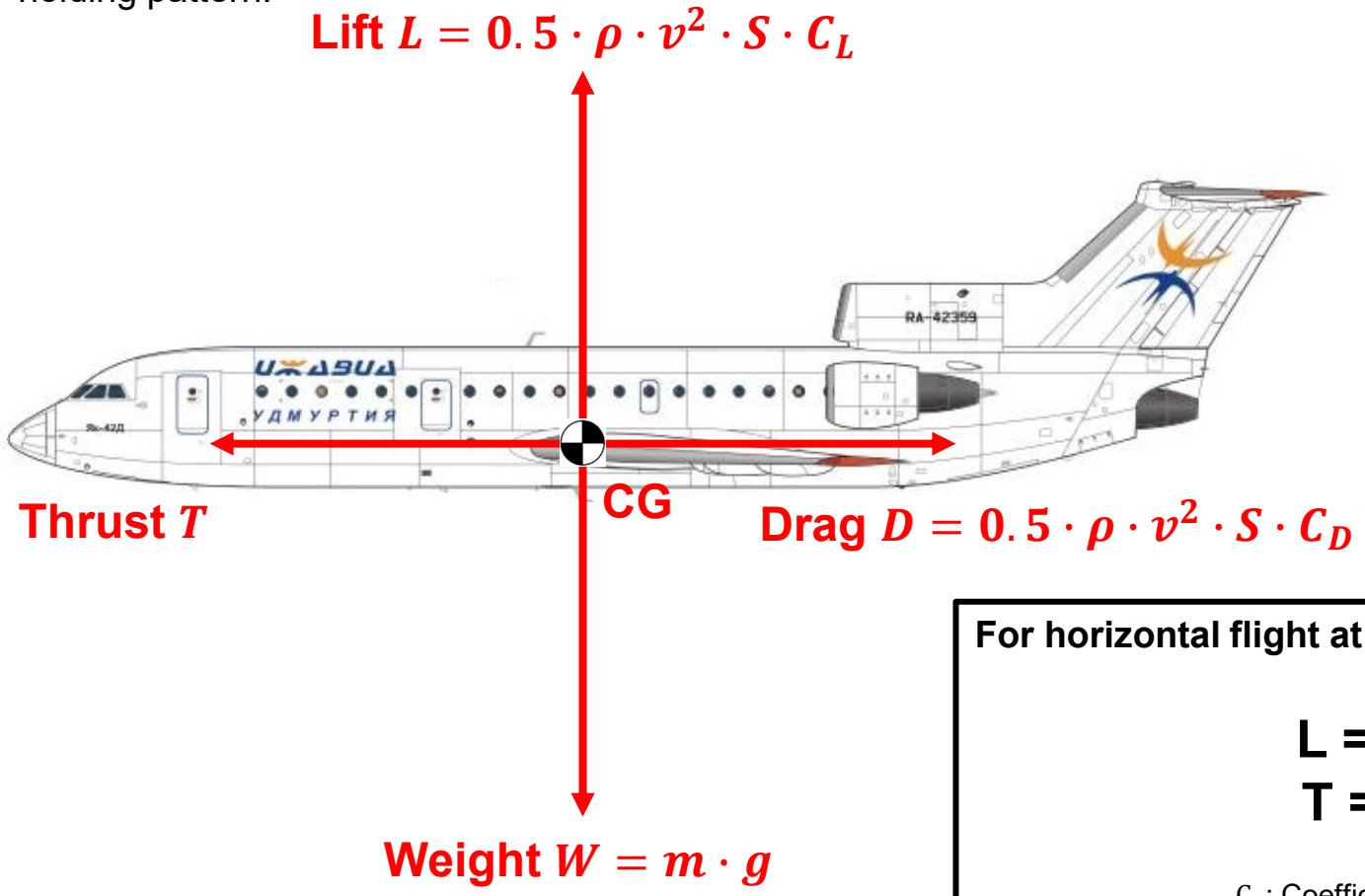
Boeing 767-300ER



Source: Boeing

Mission Analysis: The Payload-Range Diagram – Lift & Drag

Calculation for horizontal unaccelerated flight without angle of attack ($\alpha = 0°$) – estimation for cruise and holding pattern:



For horizontal flight at constant speed:

L = W
T = D

C_L: Coefficient of Lift
C_D: Coefficient of Drag
S: Wing area

Mission Analysis: The Payload-Range Diagram – Breguet Equation

$$dm_{fuel} = SFC \cdot T \cdot dt$$

$$v = \frac{dx}{dt} \Rightarrow dt = \frac{1}{v} \cdot dx$$

$$dm_{fuel} = SFC \cdot \frac{m \cdot g}{L/D} \cdot \frac{1}{v} \cdot dx$$

$$\frac{L}{D} = \frac{L}{D} \Rightarrow D = \frac{L}{L/D} = \frac{m \cdot g}{L/D} = T$$

$$dx = \frac{L/D \cdot v}{SFC \cdot g} \cdot \frac{1}{m} \cdot dm_{fuel}$$

$$dm = -dm_{fuel}$$

$$m_2 = m_1 - m_{fuel}$$

$$\int_0^R dx = -\frac{L/D \cdot v}{SFC \cdot g} \cdot \int_{m_1}^{m_2} \frac{dm}{m} = R = -\frac{L/D \cdot v}{SFC \cdot g} \cdot \ln m \Big|_{m_1}^{m_2} = -\frac{L/D \cdot v}{SFC \cdot g} \cdot \ln \frac{m_2}{m_1}$$

$$R = -\frac{L/D \cdot v}{SFC \cdot g} \cdot \ln\left(\frac{m_1 - m_{fuel}}{m_1}\right)$$

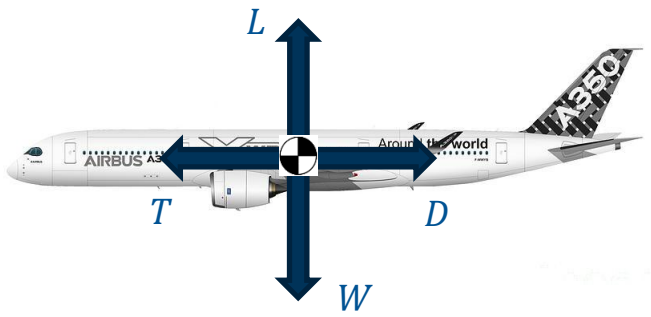
$$R = -\frac{L}{D} \cdot \frac{v}{g} \cdot \frac{1}{SFC} \cdot \ln\left(1 - \frac{m_{fuel}}{m_1}\right)$$

Aerodynamic Efficiency | **Propulsive Efficiency** | **Structural Efficiency**

m	A/C mass
m_1	A/C mass at Top-Of-Climb
m_2	A/C mass at Top-Of-Descent
m_{fuel}	Fuel mass
SFC	Specific Fuel Consumption = f(H,v)
T	Thrust = f(H,v)
t	Time
v	Speed
x	Portion of flight route
g	Gravitational constant ; 9.81 m/s2
L	Lift
D	Drag = f(C_L)
L/D	Aerodynamic efficiency = f(H;C_L)
R	Range

Exercise: Mission Calculation (1)

Mechanical equilibrium



How much **fuel** does an Airbus A350-1000 need for a **cruise flight**: Singapore – Taipei?

Coefficient of lift at cruising altitude $C_{L,C}$?
Mass at top of decent $\boldsymbol{m_2}$**: 227.6 t** (incl.: Contingency fuel: 15 min, Fuel to alternate airport, Final reserve fuel: 30 min, Fuel for decend & landing included)
Density at top of decent $\boldsymbol{\rho_2 = 0.38\ \frac{kg}{m^3}}$ (ref. FL350)
Cruising **speed** $\boldsymbol{v_c}$: $\mathbf{252\ \frac{m}{s}}$ (Mach M 0.85)
Reference **wing area** $\boldsymbol{S}$: $\mathbf{370\ m^2}$

$$L = m_2 \cdot g = \frac{\rho_2}{2} \cdot v_c^2 \cdot S \cdot C_{L,C}$$

$$C_{L,C} = 0.500$$

Picture source: fineartamerica.com; engineering.com

Exercise: Mission Calculation (2)

Cruise flight at **FL350 (35000 ft)**

Distance for cruise R_c: **2361 km**

Mass at top of decent m_2: **227.6 t**

Cruising **speed** v_c: $252\ \frac{\mathrm{m}}{\mathrm{s}}$ (Mach M 0.85)

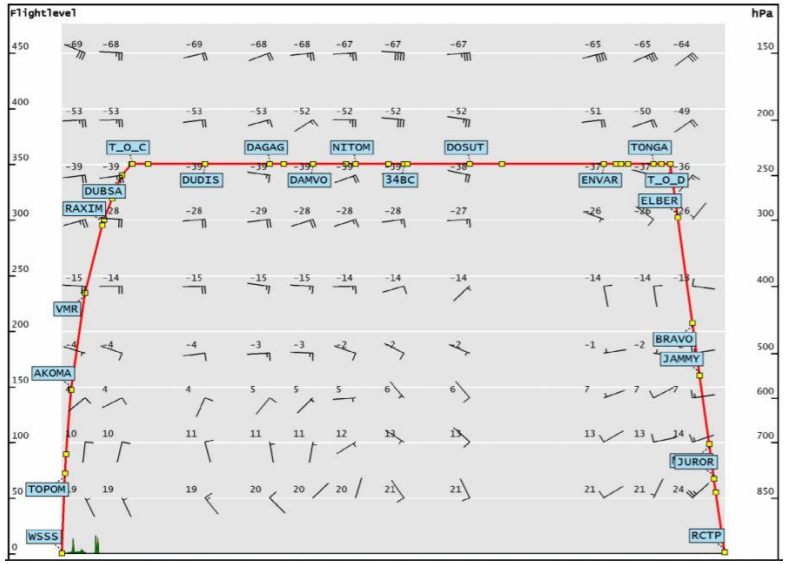
Coefficient of drag $C_{D,C}$: 0.03

Specific fuel consumption: SFC: $1.2 \cdot 10^{-5}\ \frac{kg}{N \cdot s}$

Density fuel ρ_f: $0.785\ \frac{kg}{l}$

Passengers onboard n_{PAX}: **279 PAX** (LF = 85%, 328 seats)

Climb, cruise and decent path of route: Singapore-Taipeh (reality)



Breguet:

$$m_1 = \exp\left(\frac{R_c}{v_c} \cdot g \cdot SFC \cdot \frac{C_{D,c}}{C_{L,c}}\right) \cdot m_2 = 243.2\ t$$

$$m_1 - m_2 = 15.6\ t$$

- Fuel consumption

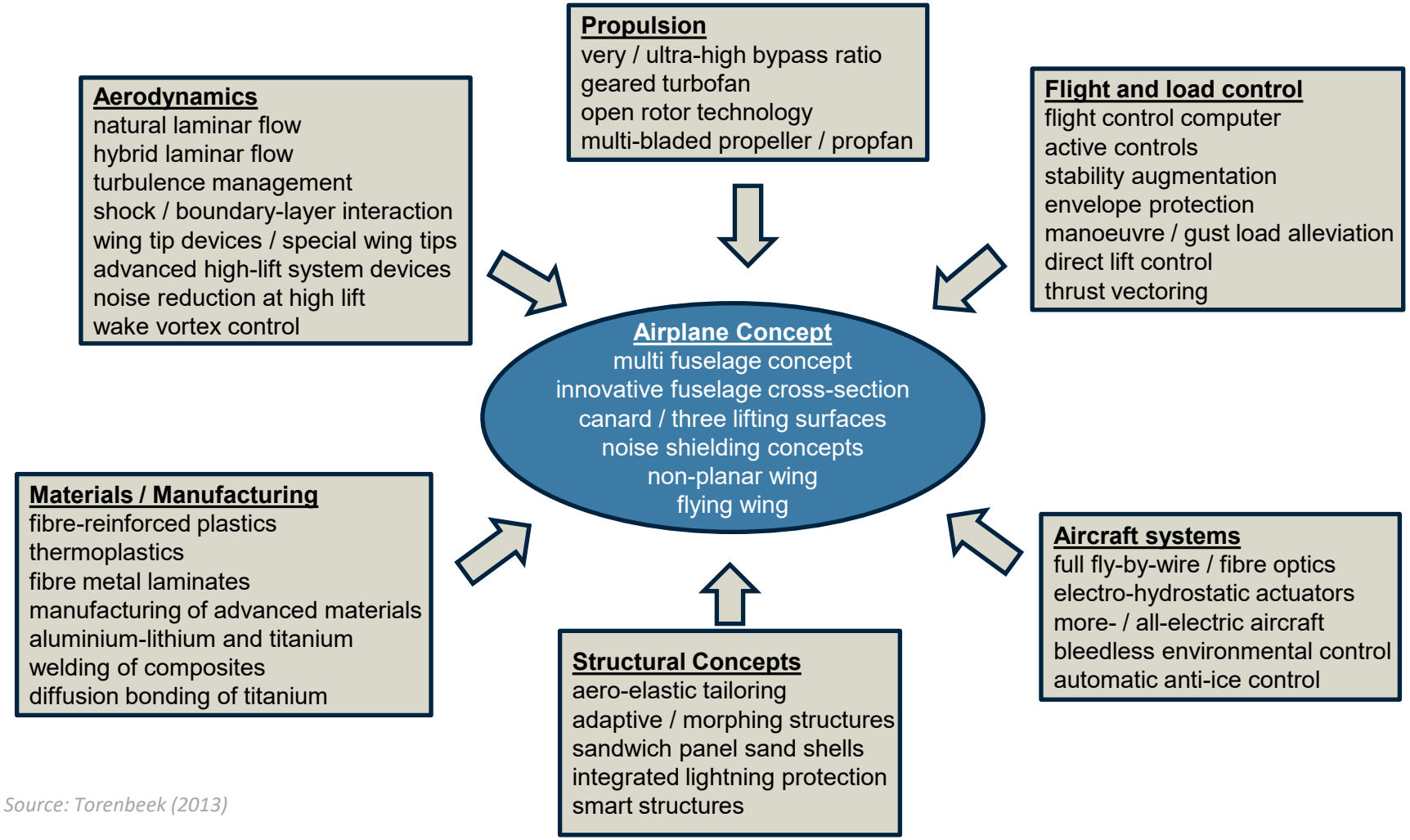
$$V_f = \frac{m_2 - m_1}{\rho_f} = 19836\ l$$

$$\frac{V_f \cdot 100}{n_{PAX} \cdot R_c} = 3.0\ \frac{l}{100 PAX km}$$

Picture source: simbrief.com

Disciplines and Enabling Technologies

Conceptual design studies are based on choices from the most promising advancements in the state of the art.



Source: Torenbeek (2013)

Aircraft Commonality

Aircraft commonality plays a significant role when an airline selects a new type of aircraft.

Commonality concerns different aircraft components:

- Propulsion system
- Flight deck: *Cross Crew Qualification (CCQ)*
- Aircraft components (e.g., fuselage sections, wing)
- Various aircraft sub-systems

The positive effects of aircraft commonality grow with an increasing size of the aircraft fleet.

Aircraft commonality does not directly affect Aircraft Direct Operating Costs (DOC). The advantages of aircraft commonality can only be determined for each airline on an individual basis.

ΔC = f(type of fleet, size of fleet, etc.)

Aircraft Commonality: Cost Saving Potentials

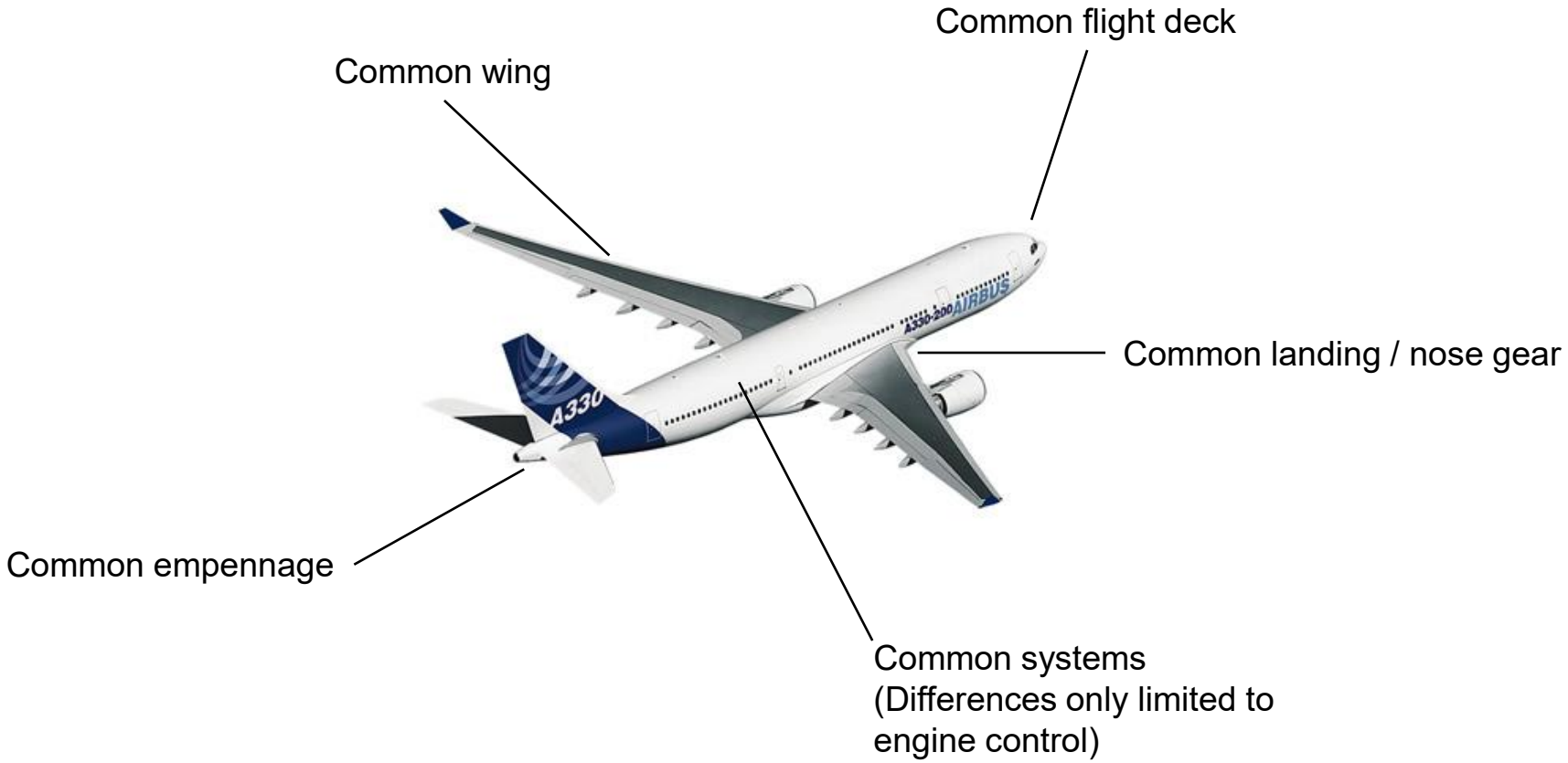
Primary saving potentials:

- Reduction of pilot training hours and costs
- Reduction of number and utilization hours of flight simulators
- Increased productivity of pilots (flexibly usable, CCQ)
- Reduction of spare parts storage for engines, aircraft cell, sub-systems
- Reduction of maintenance costs

Secondary saving potentials:

- Reduction of administrative costs for pilot assignment
- Reduction of training expenditures for maintenance personnel
- Reduction of training expenditures for cabin personnel

Aircraft Commonality: Airbus A330 / A340



Source: Airbus

Aircraft Commonality: Aircraft Family

2-class seat config.

186

A321

+6,93 meters

+13 frames

150

A320

Baseline

124

A319

-3,73 meters

-7 frames

107

A318

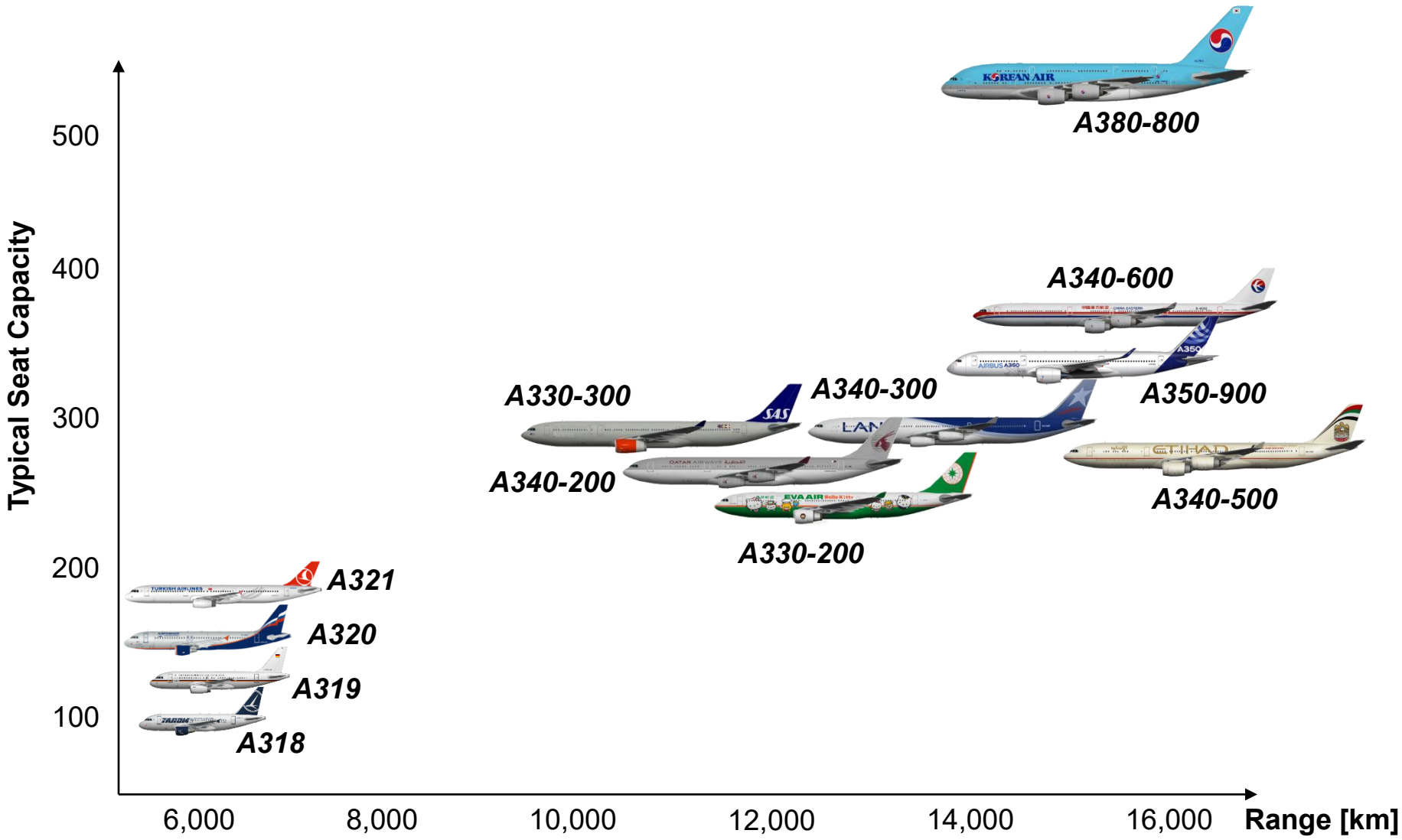
-6,13 meters

-11,5 frames

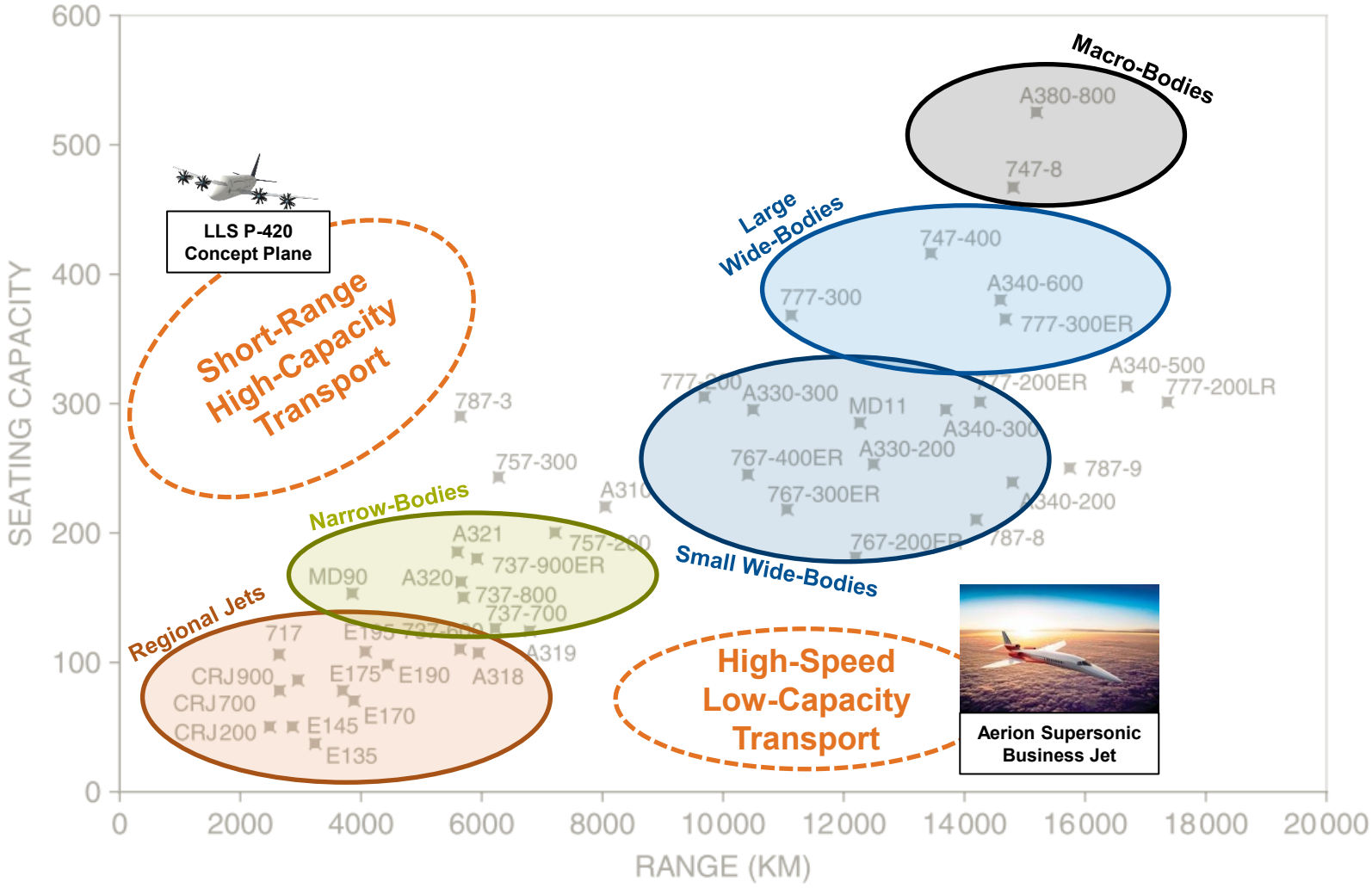


Source: Airbus

Aircraft Commonality: Airbus Product Families



Manufacturers' Product Policies

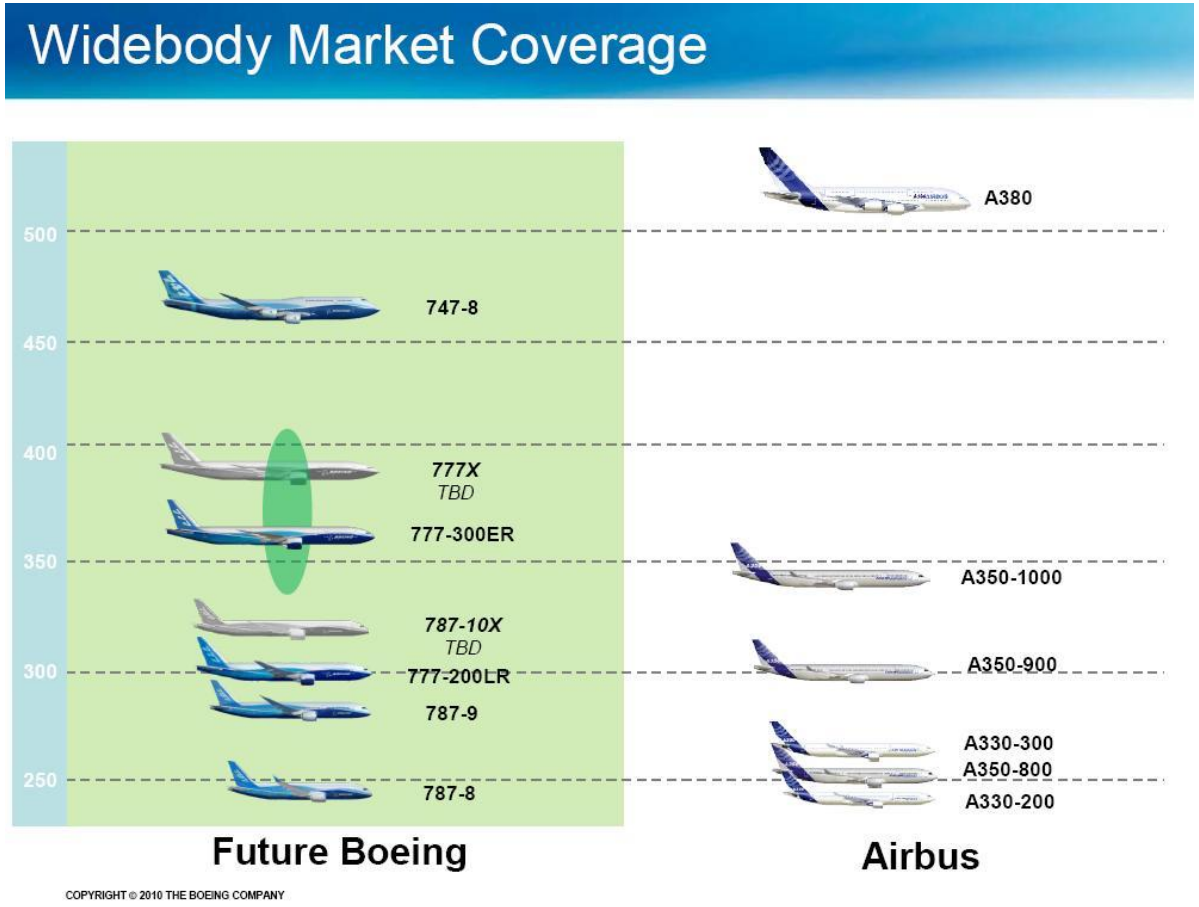


Airbus vs. Boeing on Wide-Bodies

Airbus View



Boeing View



Technology Transfer: Airbus A330 / A340

Newly designed wing

High commonality to existing types
(A300 / A310 / A320)



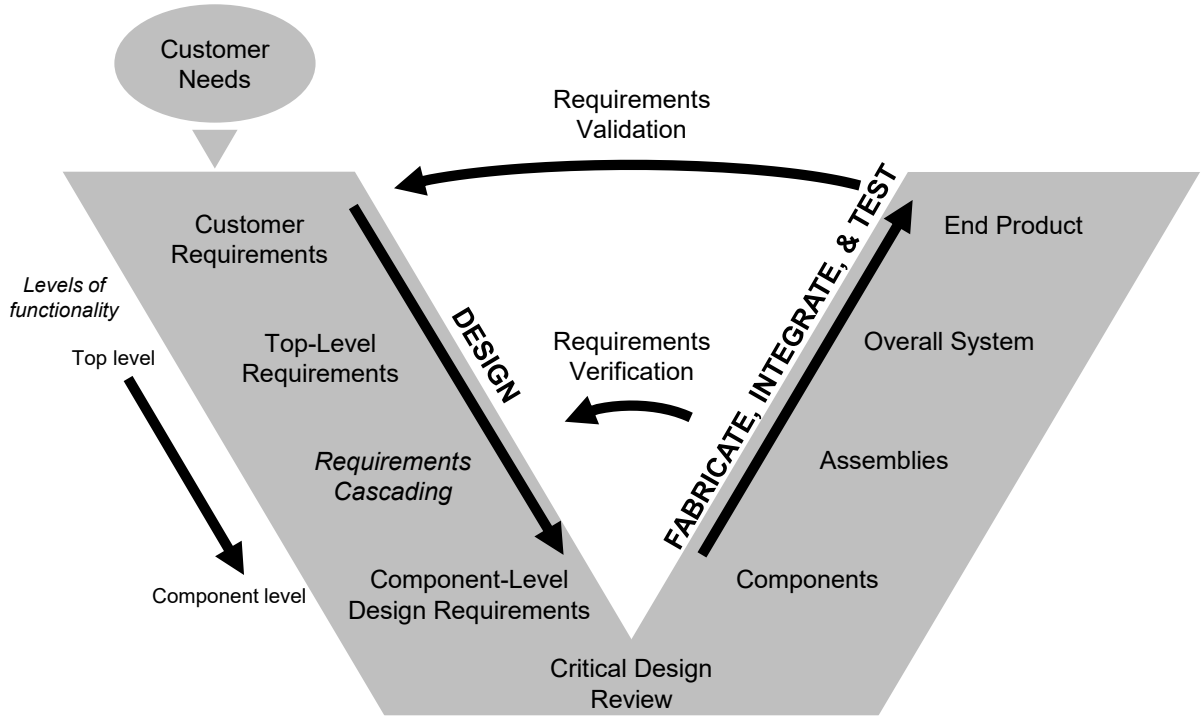
Transfer from A320 (relevant experience)
- System technology
- Flight guidance and control
- Design of flight deck
- Integration of engines
- Use of composite materials

Transfer from A310 / A300-600
- Fuselage cross section
- Trim tank
- Cargo loading systems
- Use of composite materials
- Empennage

Source: Airbus

Requirements Engineering

The V-Model



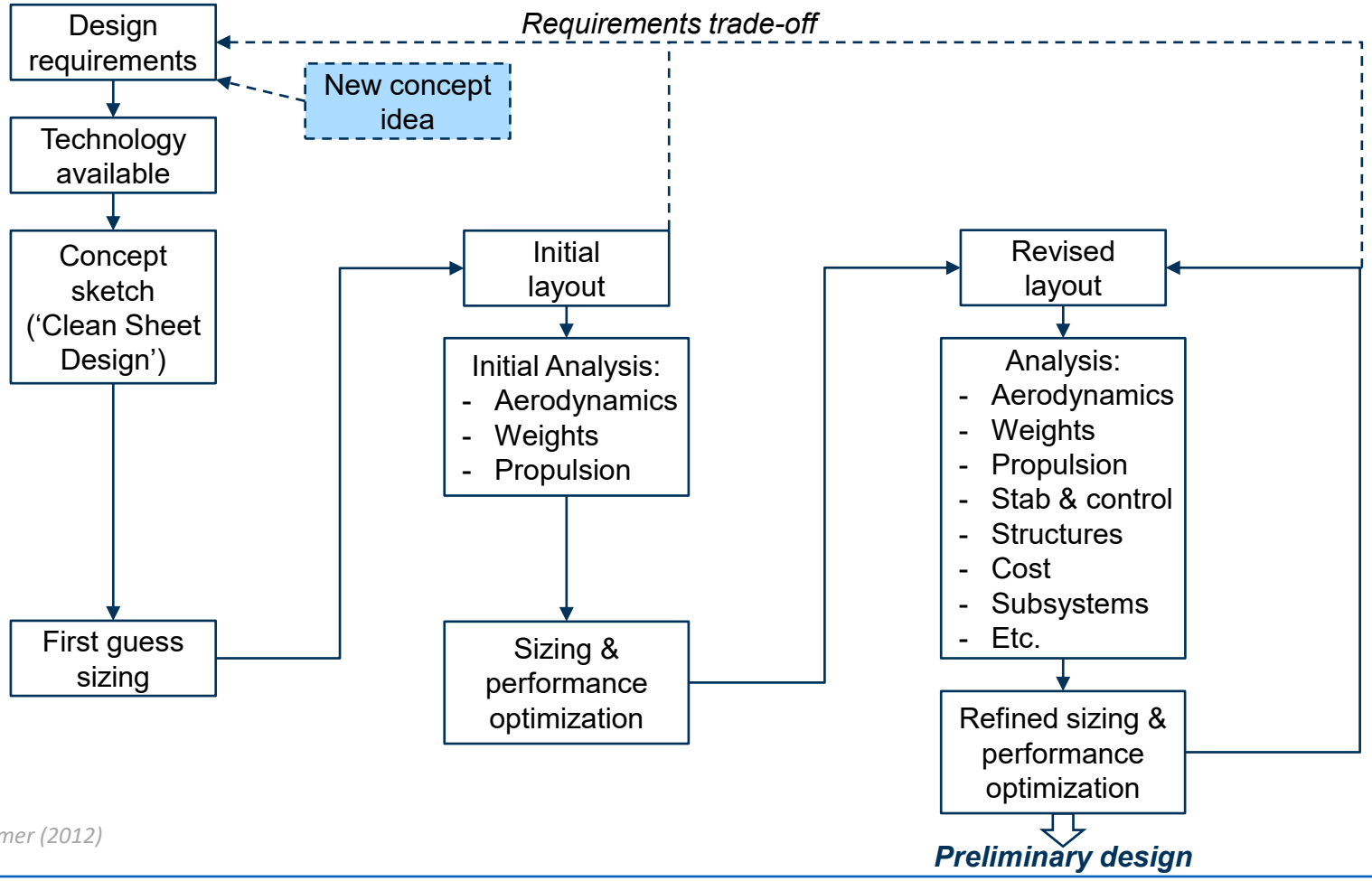
Validation is the process of confirming, “through the provision of objective evidence, that the requirements for a specific intended use or application have been fulfilled.”

Verification is the process of confirming, “through the provision of objective evidence, that specified requirements have been fulfilled.”

(ISO 9000)

Aircraft Conceptual Design (1)

Aircraft conceptual design is an iterative process of requirements trade-off and design contraints.



Source: Raymer (2012)

Aircraft Conceptual Design (2)

The goal of the **mission analysis** is to identify and describe the different phases of the flight mission that the aircraft will have to accomplish according to its specification. This comprises both the on-ground (taxi operations, take-off, and landing) and in-flight (climb, cruise, descent, loiter) mission segments.

The **initial sizing** (also called ‘first-guess sizing’) is aimed at making a first assumption on the overall weight of the aircraft as a function of the payload, range, and speed requirements. Usually, this task is accomplished using statistical data of existing (and retired) aircraft.

During the **parametric sizing** phase, the initial aircraft concept is refined by trading off the thrust, weight, and wing-related characteristics of the aircraft against each other with the goal to determine the optimal engine thrust and wing area, which will eventually lead to a revised concept > use of a **constraint diagram**.

Source: Price and Soban (2010)

Aircraft Conceptual Design (3)

Subsequently, several **configuration options** have to be created that express how the different components of the aircraft (e.g., wing, fuselage, engines) can be arranged.

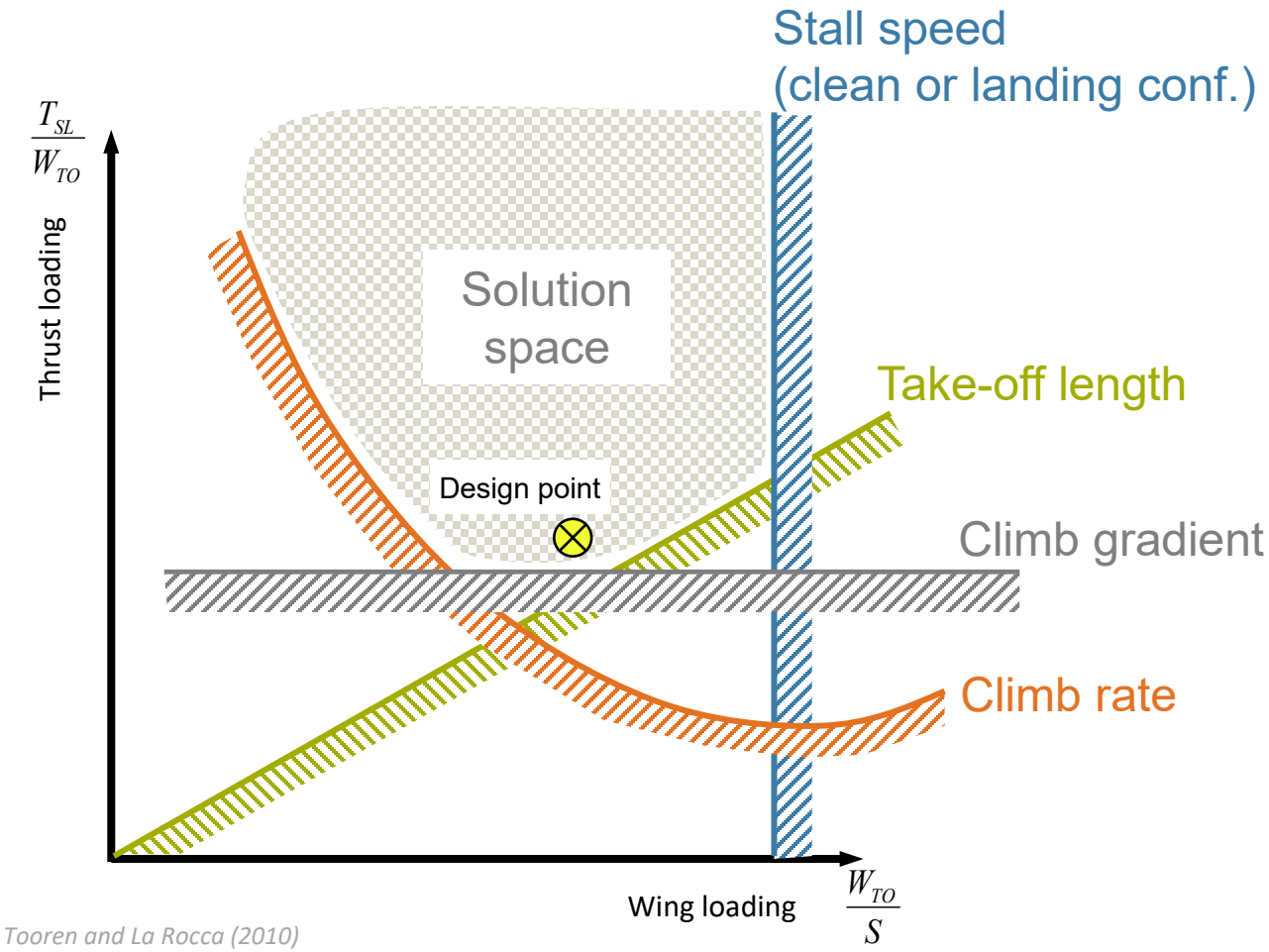
Having selected one aircraft configuration, the empty weight of this configuration is estimated during the **weight estimation** phase. In opposition to the initial sizing task, however, more precise weight estimation methods can be applied at this stage, as the actual aircraft configuration has been set.

In conclusion, the conceptual design process provides an estimate of the weight, size, and configuration of the aircraft, and some more detailed information such as engine size, specific fuel consumption, and desired performance characteristics for the different mission segments.

Source: Price and Soban (2010)

Aircraft Conceptual Design (4)

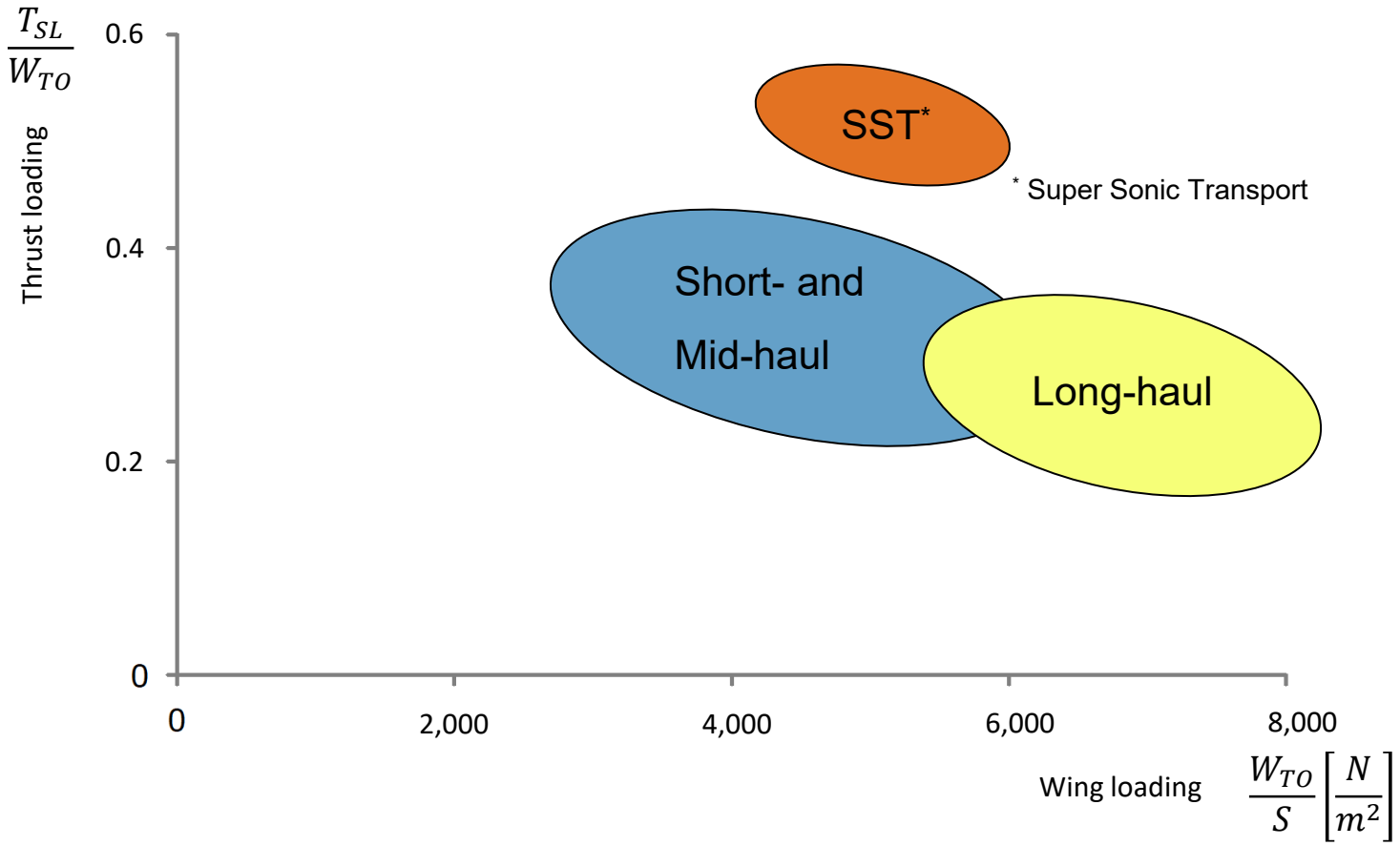
Constrained thrust-loading/wing loading design space for a civil transport aircraft



Source: van Tooren and La Rocca (2010)

Aircraft Conceptual Design (5)

Thrust-loading/wing loading design space for civil transport aircraft



Aircraft Conceptual Design

Key Messages

Aircraft Conceptual Design is

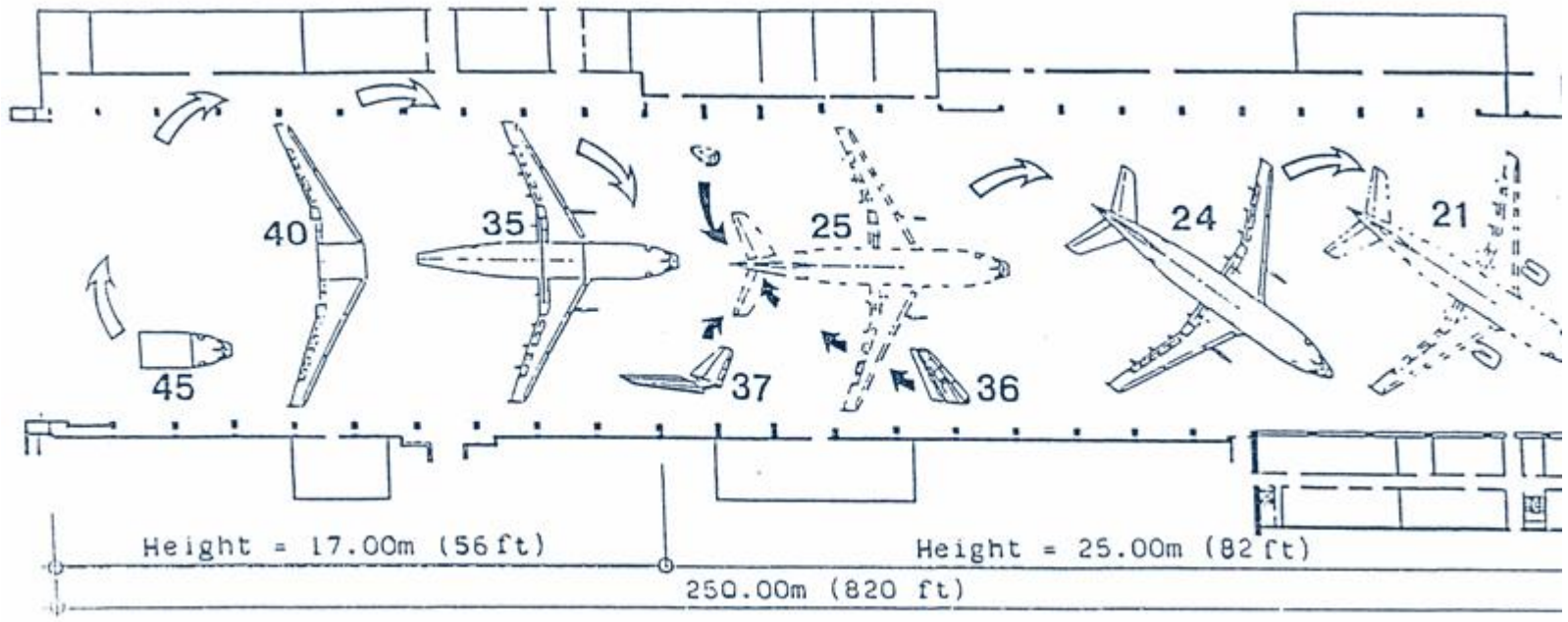
- … a highly iterative process
- … is based on statistical and empirical functions
- … relies on data of aircraft already flying
- … does not automatically deliver the *best* preliminary design

→ An aircraft concept

- … a ‘big compromise’
- … is the result of a feasibility study
- … is a conceptual study for the analysis of technological potentials
- … inheres a lot of uncertainty
- … forms the basis for further work

Prior to proceeding with the aircraft preliminary design stage, a comprehensive concept assessment is required.

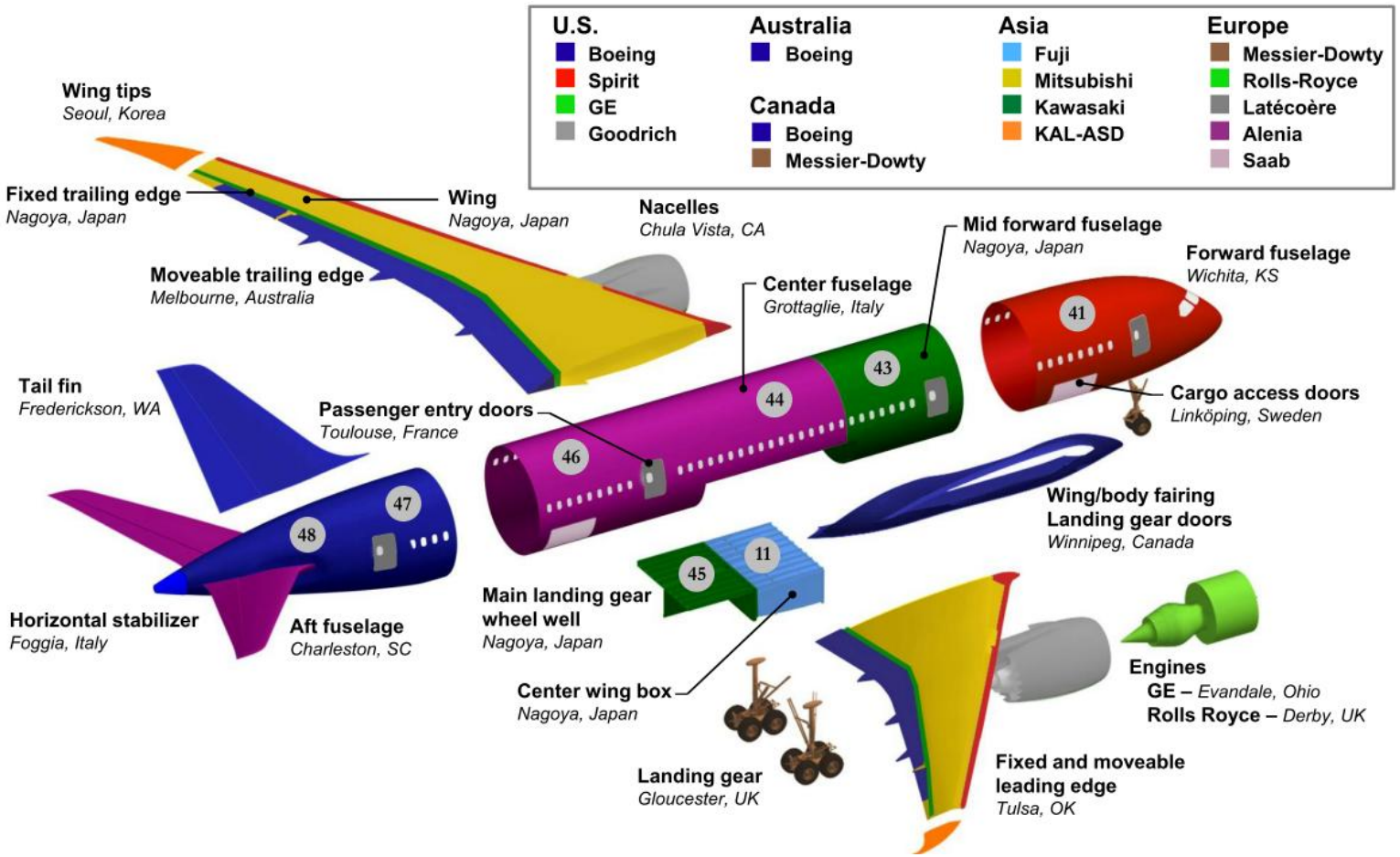
Aircraft Production



Station 45:	Assembly of forward fuselage; electrical and mechanical equipment installation
Station 40:	Assembly of wing and center fuselage
Station 36/37:	Installation of vertical stabilizer and horizontal stabilizer
Station 35:	Fuselage assembly sections #11 to #19, pylons, landing gear
Station 25:	Horizontal tailplane, vertical stabilizer, tail cone, furnishing and installation
Station 24:	Final installation, fuel tank sealing test
Station 21:	Powerplant installation and general tests, neutral station

Source: Airbus

Fragmentation of production: Boeing 787 Dreamliner



Source: Boeing

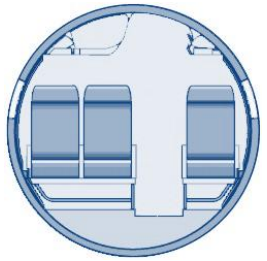
3. Aircraft Life Cycle and Development

3.4 Aircraft Cabin Design



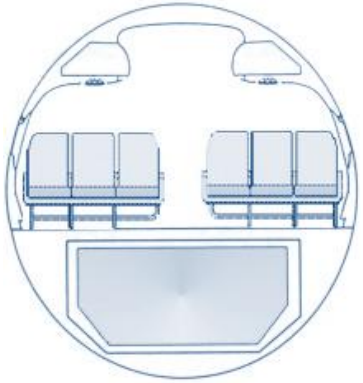
Fuselage Cross-Sections

Commuter



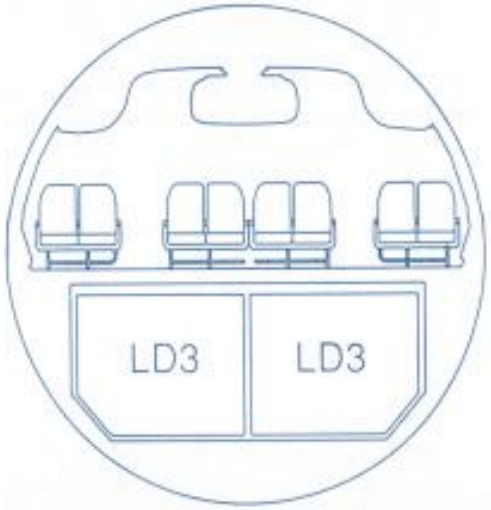
Single Aisle

Narrow Body



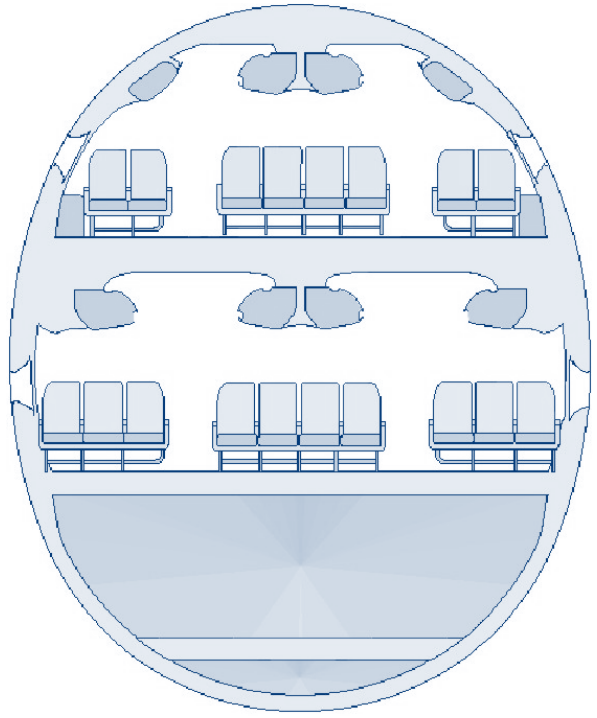
Single Aisle

Wide Body



Twin Aisle

Macro Body



Two Deck

Cabin Layout: Airbus A340-300

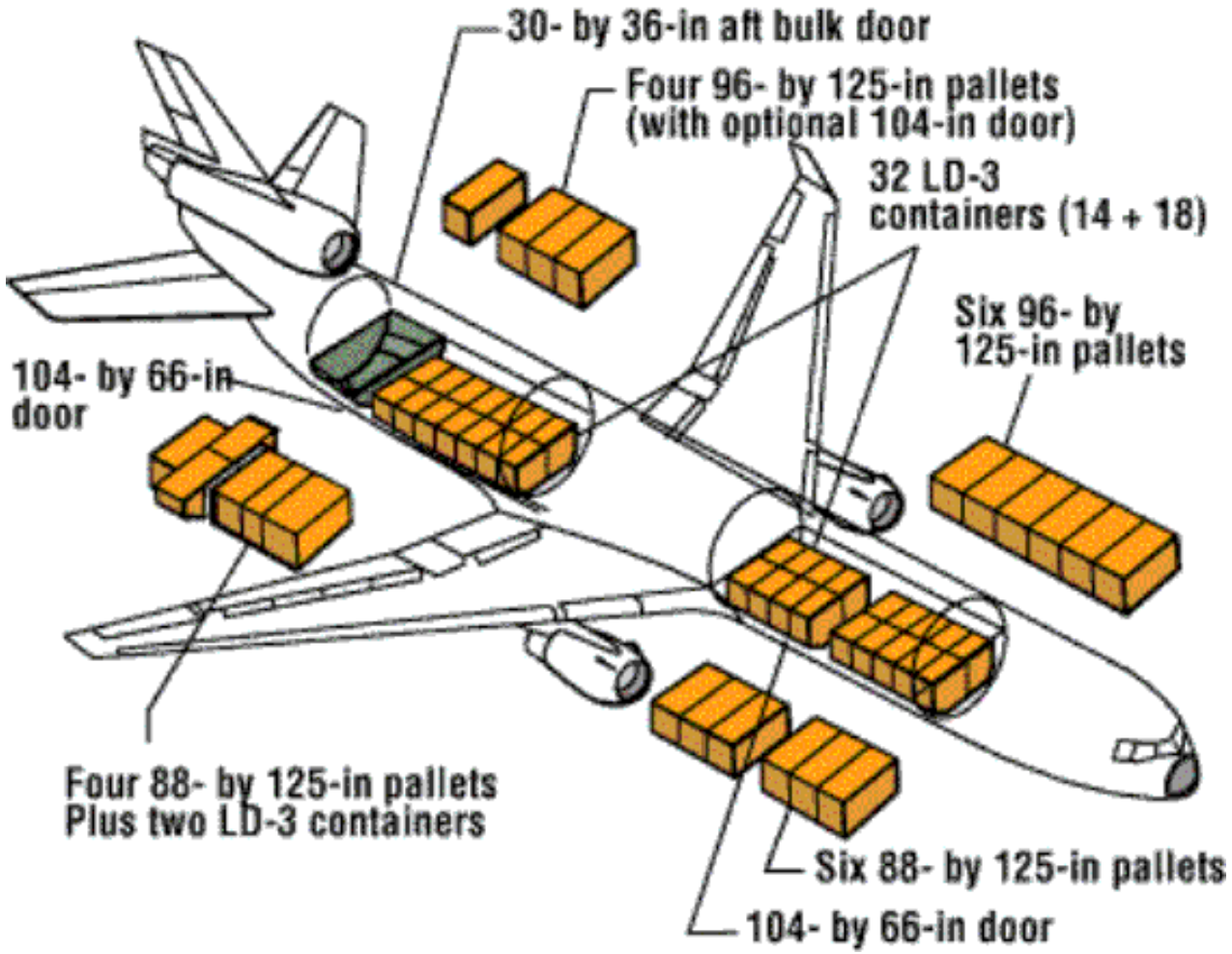


Source: Airbus

Cabin Design: Definition of Standards

		***SR* Short-Range** *SR ≤ 3000 nm*	***MR* Mid-Range** *3000 nm < MR < 5500 nm*		***LR* Long-Range** *LR ≥ 5500 nm*		
		YC	***FC***	***YC***	***FC***	***BC***	***YC***
Seats	*Seats in %* *Seat pitch [inch]* *Seat recline [inch]*	100 32 5	8 - 10 40 7.5	90 - 92 32 5	5 - 7 60 15	18 - 20 38 7	73 - 77 32 5
	Seat width (two abreast)	40	48	40	53	50	40
No. of cabin crew per Pax		1 / 45	1 / 8	1 / 35	1 / 8	1 / 20	1 / 35
Toilets per Pax		1 / 60	1 / 14	1 / 45	1 / 14	1 / 25	1 / 45
Trays per Pax		1.7	9	2.3	9	7	2.7
Storage space [inch/Pax]		n/a	1.5	n/a	1.5	1.5	n/a

Loading the MD-11



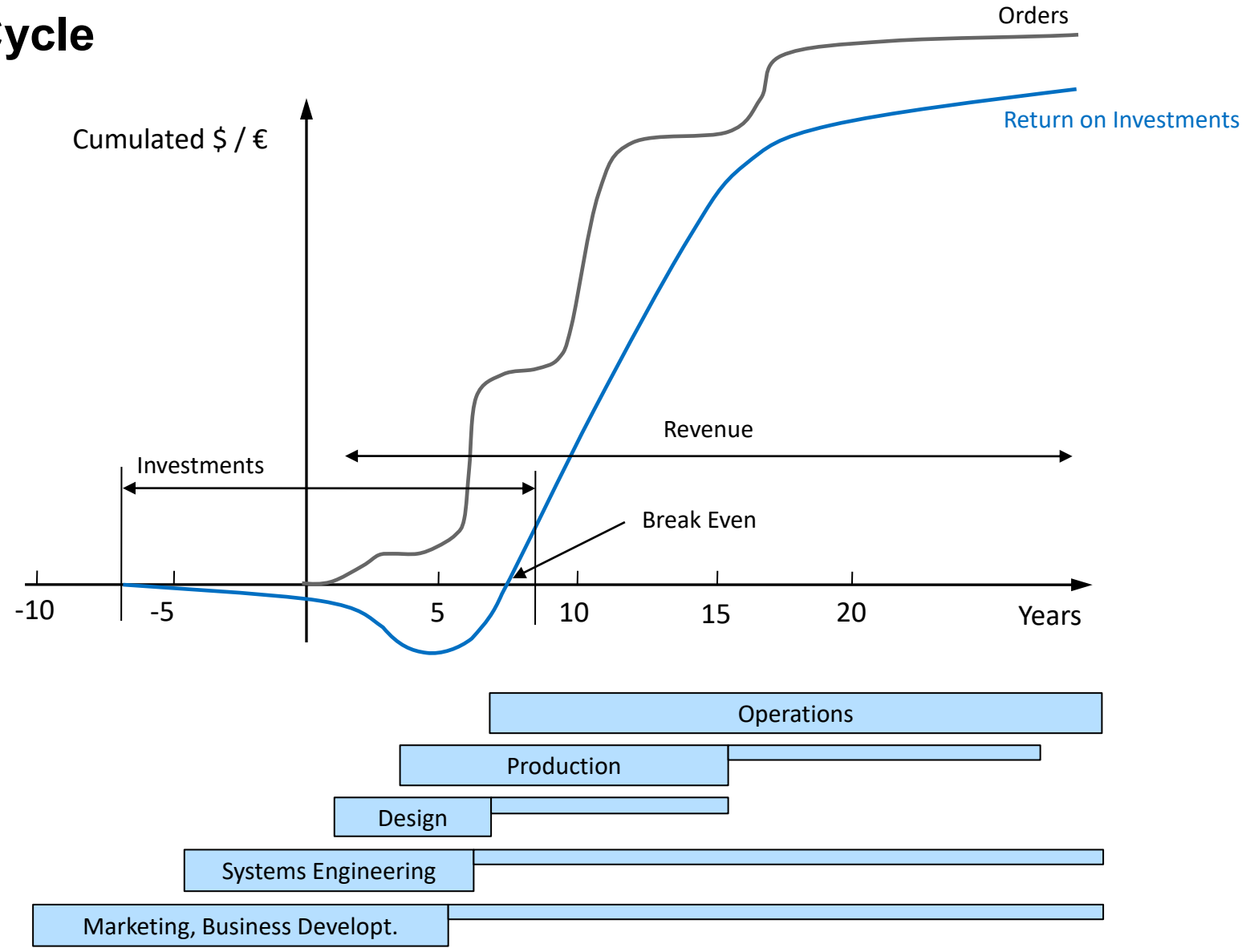
Source: Boeing

3. Aircraft Life Cycle and Development

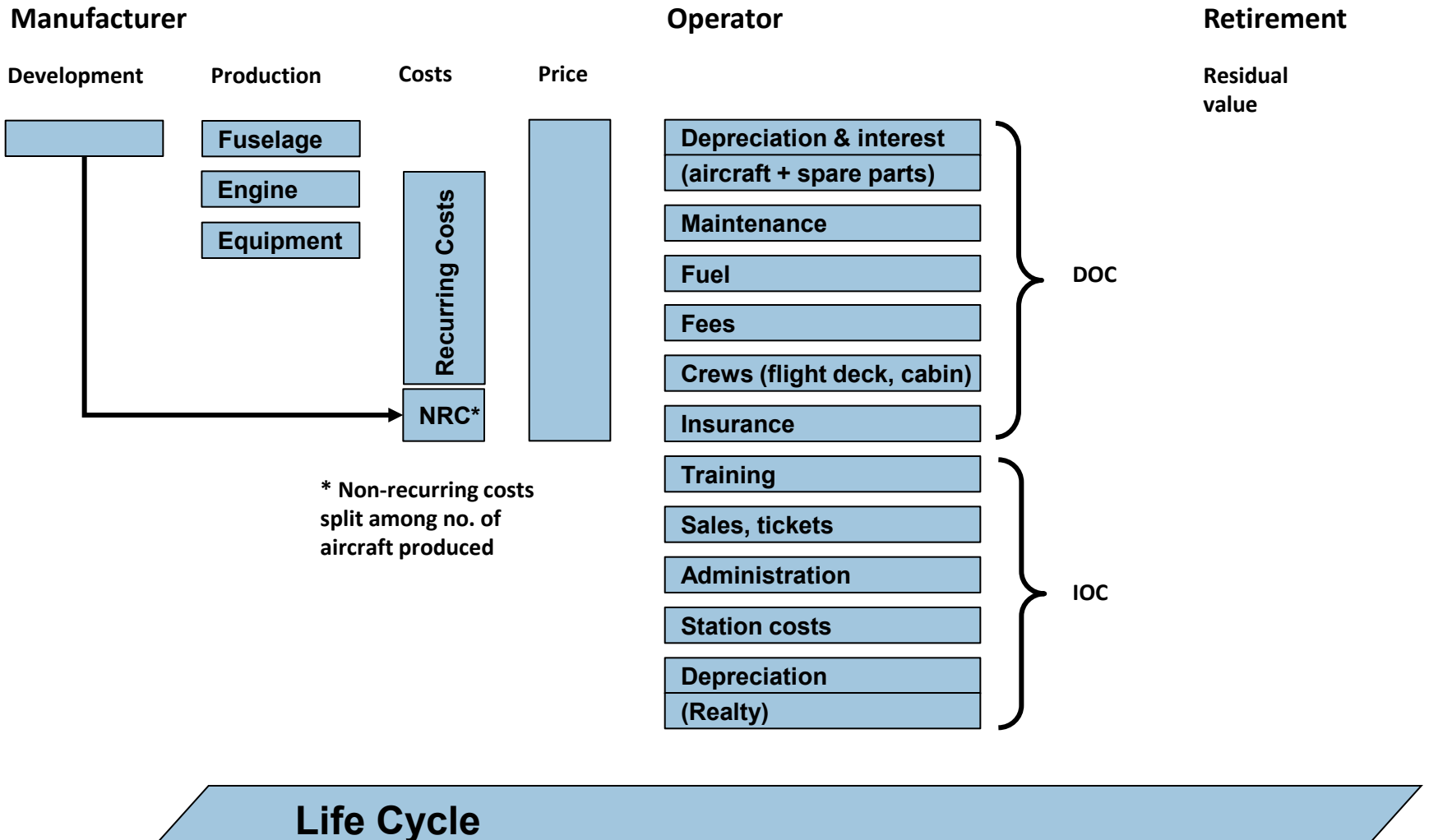
3.5 Costs



Business Life Cycle

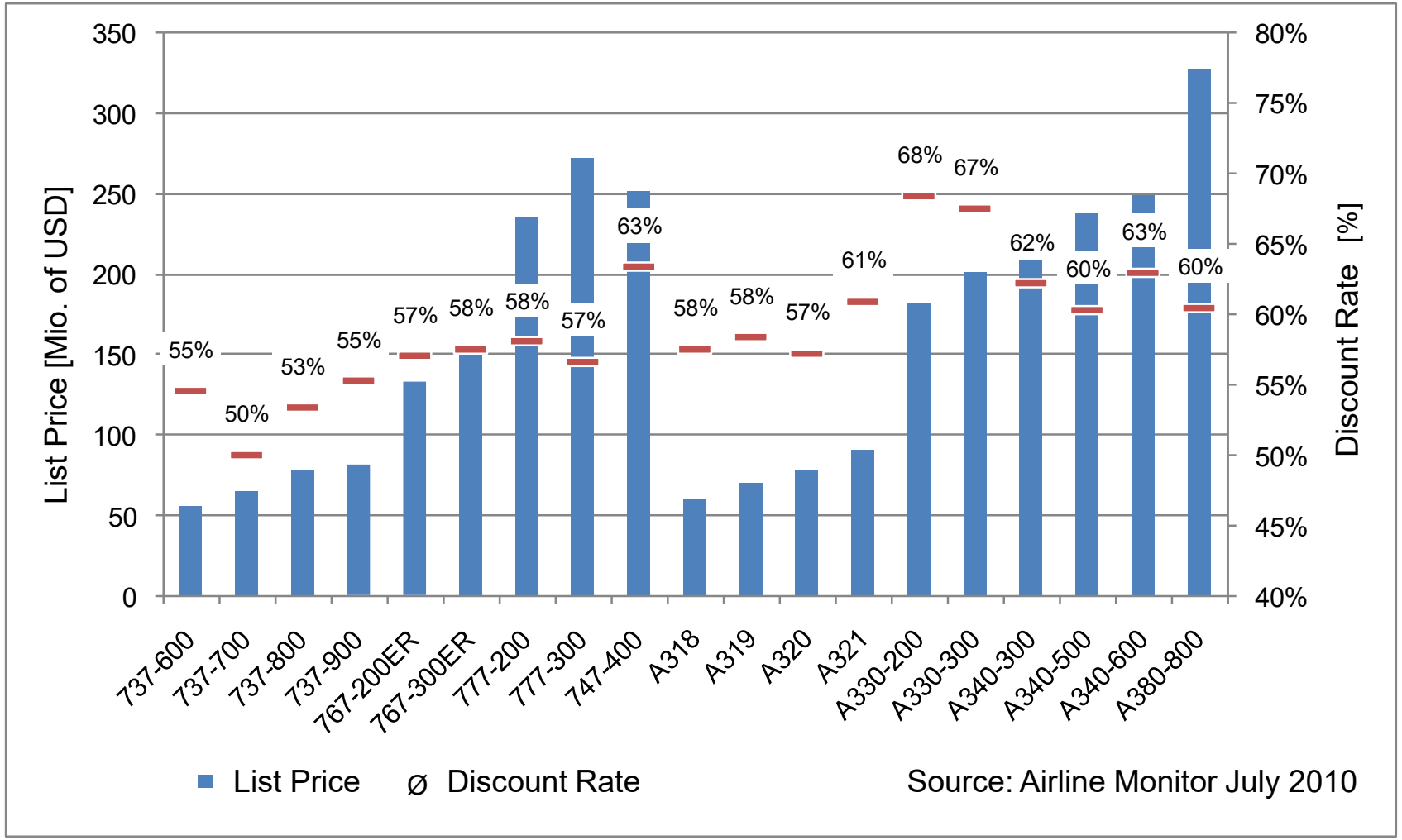


Life Cycle Costs



Source: Schäffler

Unit Prices and Discounts for Civil Aircraft (2010)



Aircraft End of Life

Two questions to be answered:

- When does an aircraft life end?
- What is the residual value?



Source: http://www.pamelalife.com/english/gallery/grues.jpg/view

Aircraft end of life:

"The period of time over which it is expected to be physically and economically feasible to operate both aircraft and engines in the intended role" [1]

End of economic life:

- Increasing operating cost exceeds revenue generating capacity
- Availability of significantly more efficient aircraft
- Environmental regulations make operations less acceptable

End of structural life:

- Repair of the structure is no longer economically feasible[2]
- Overall fatigue life usually twice as long as the minimum designed service life
- Boeing 737[3]: Service life 75,000 flight cycles, Fatigue life 150,000 flight cycle

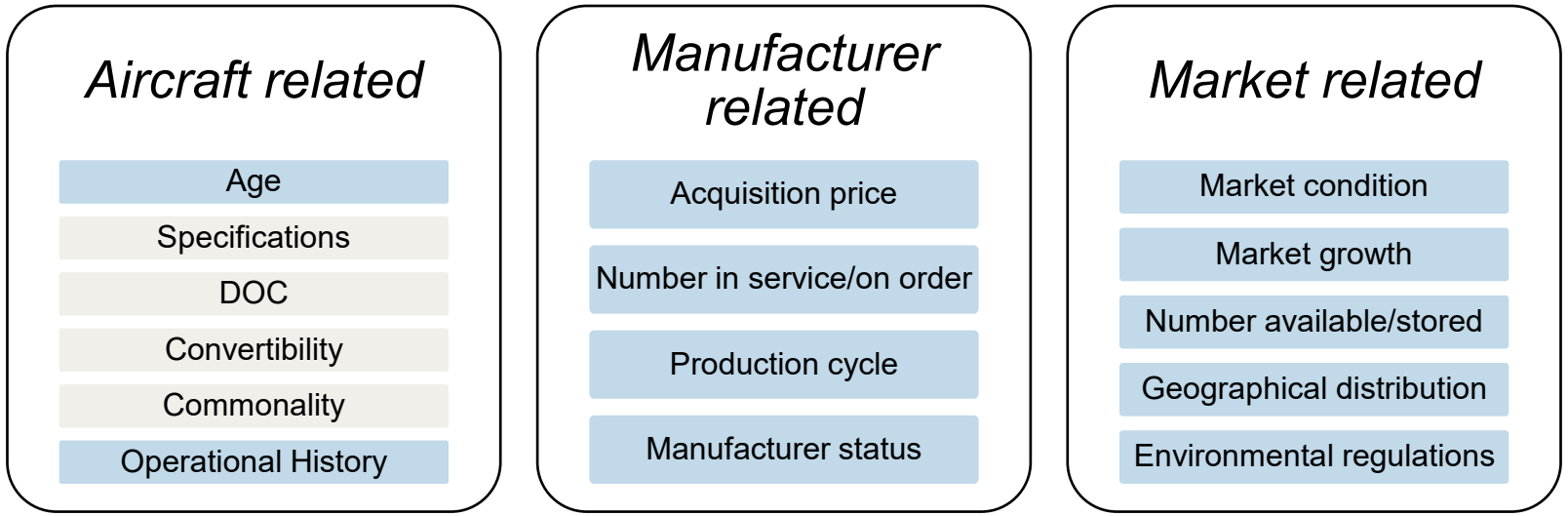
1: US Court decision, 2006

2: http://adg.stanford.edu/aa241/AircraftDesign.html

3: http://www.boeing.com/commercial/aeromagazine/aero_07/corrosn_sb_table01.html

Aircraft Residual Value

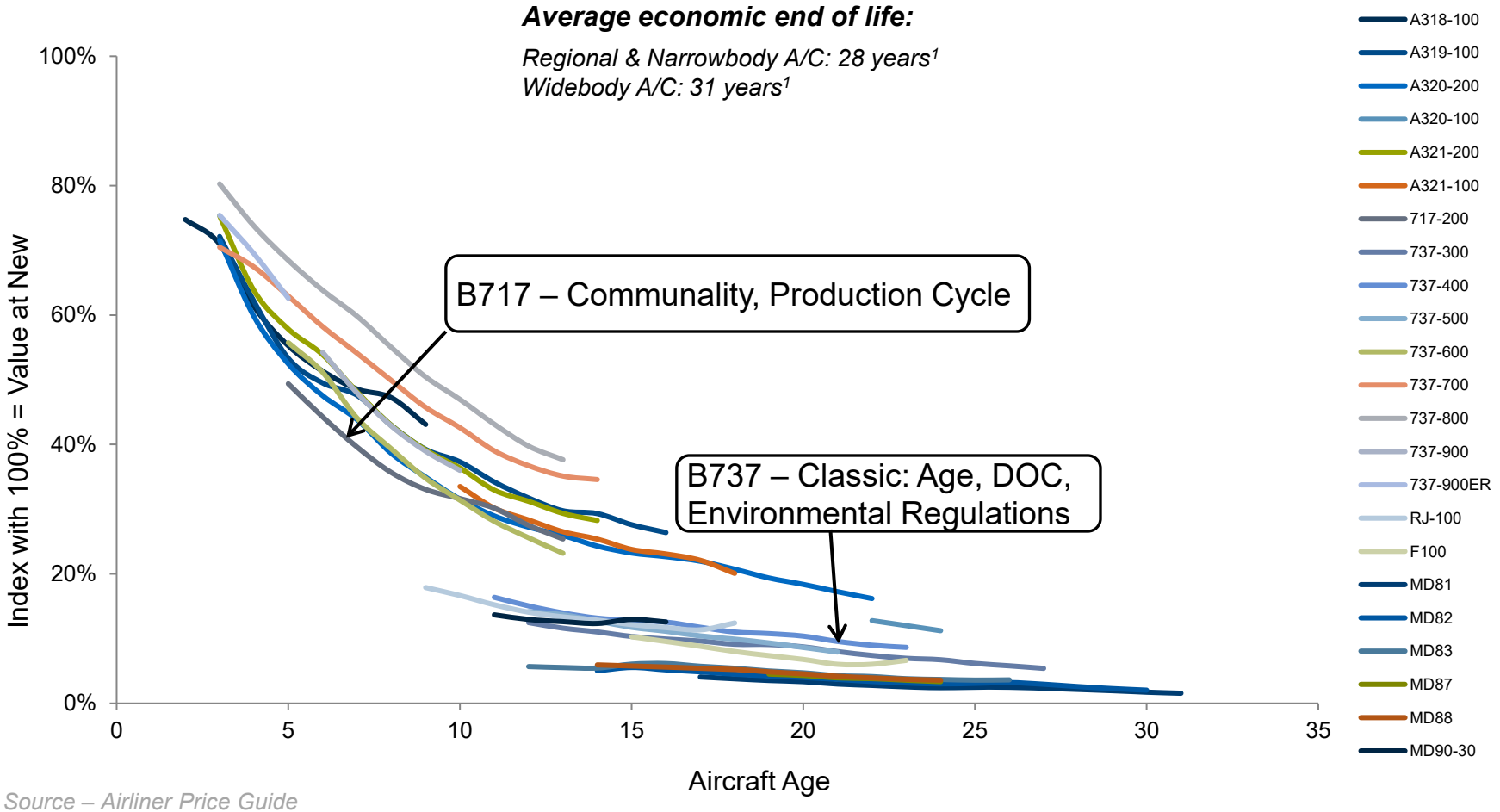
Factors influencing aircraft value[2,3]:



1: Definitions by International Society of Transport Aircraft Trading (ISTAT)
2: Kelly D. B., Forecasting aircraft values: An appraiser's perspective, Airfinance Annual
3: Ackert S., The Relationship between an aircraft's value and its maintenance status

Aircraft Residual Value

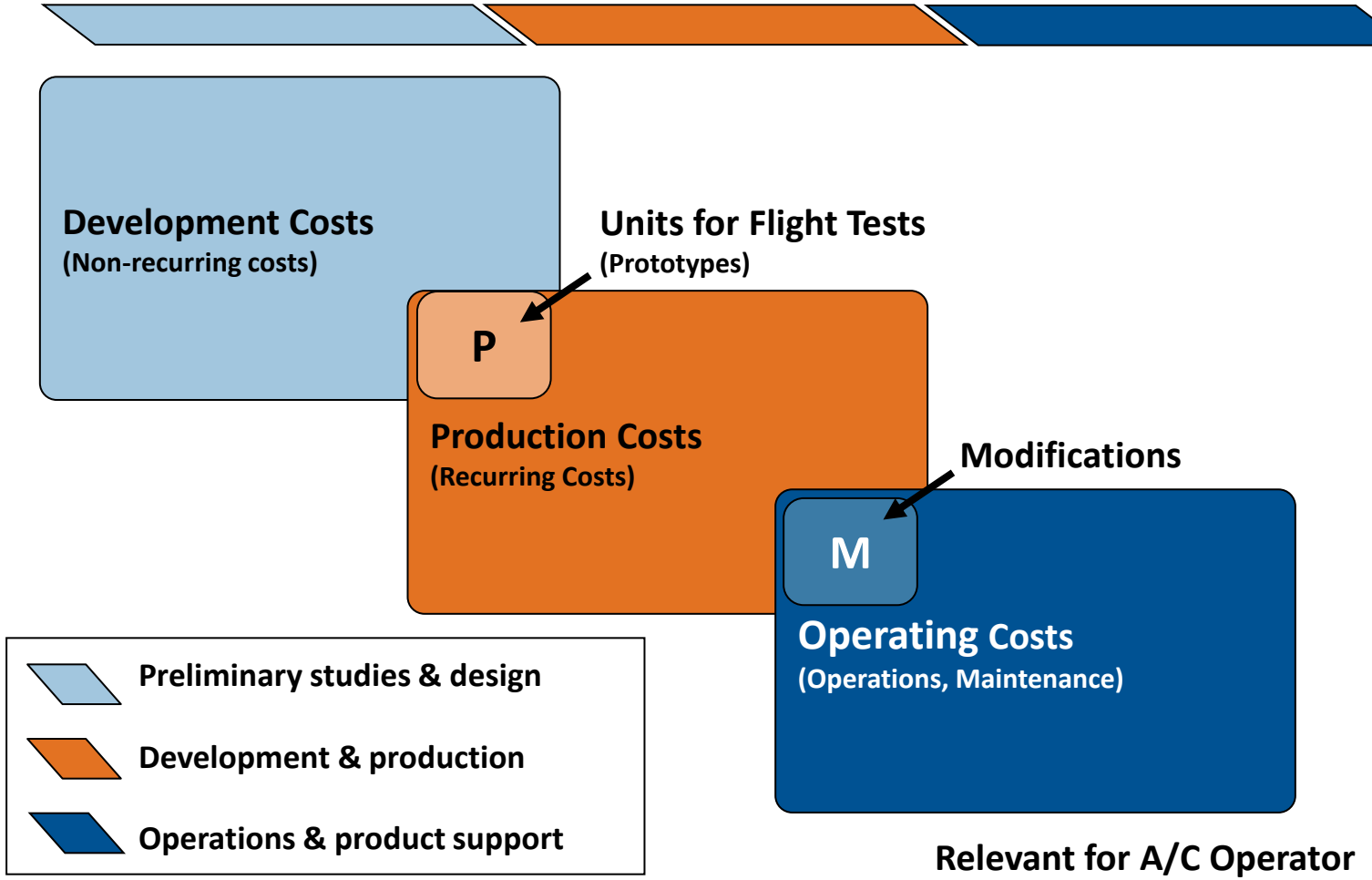
Relationship between residual value and age[1] – Narrow-body aircraft



1: Source – Airliner Price Guide

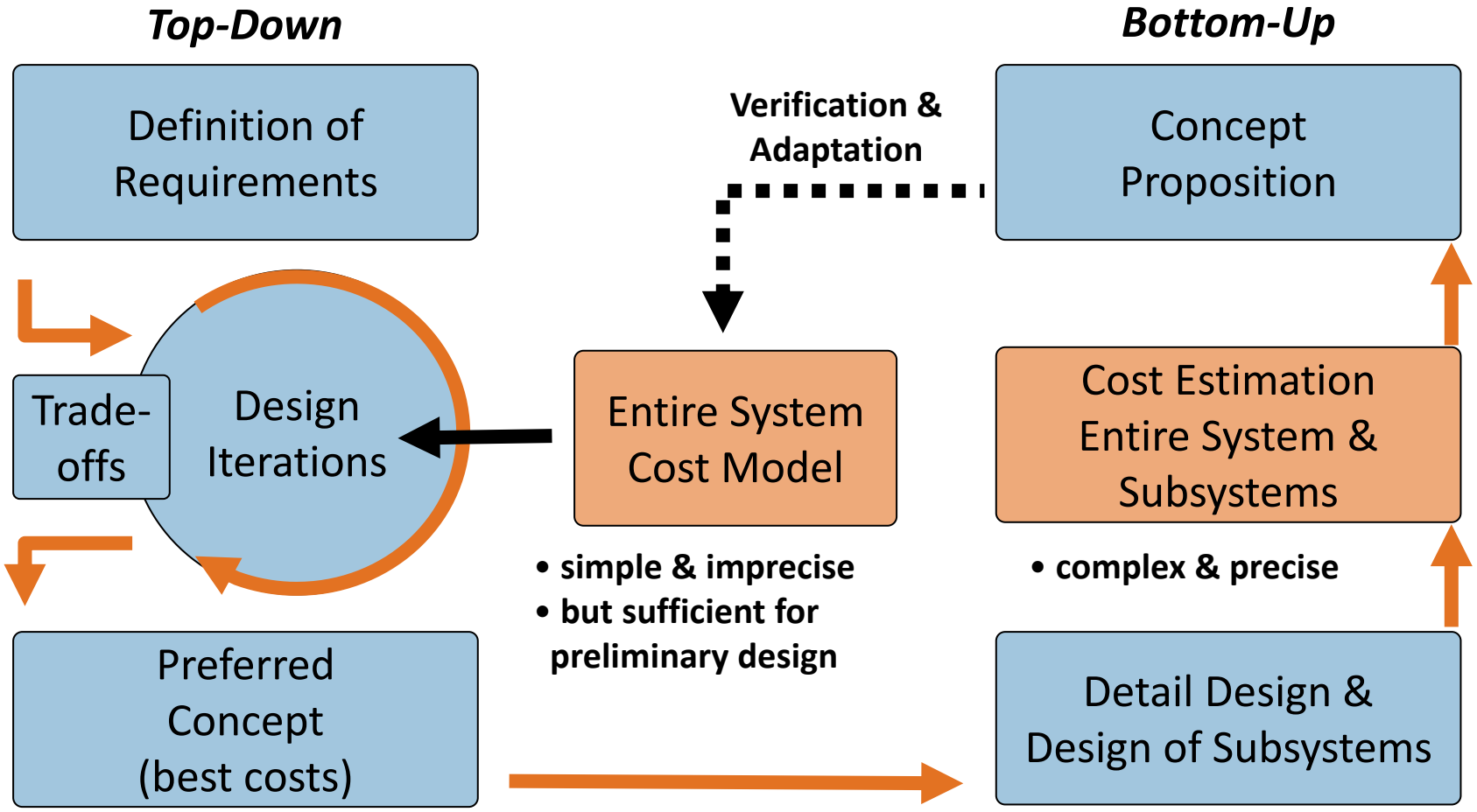
Cost Estimation

Split of costs along product life cycle

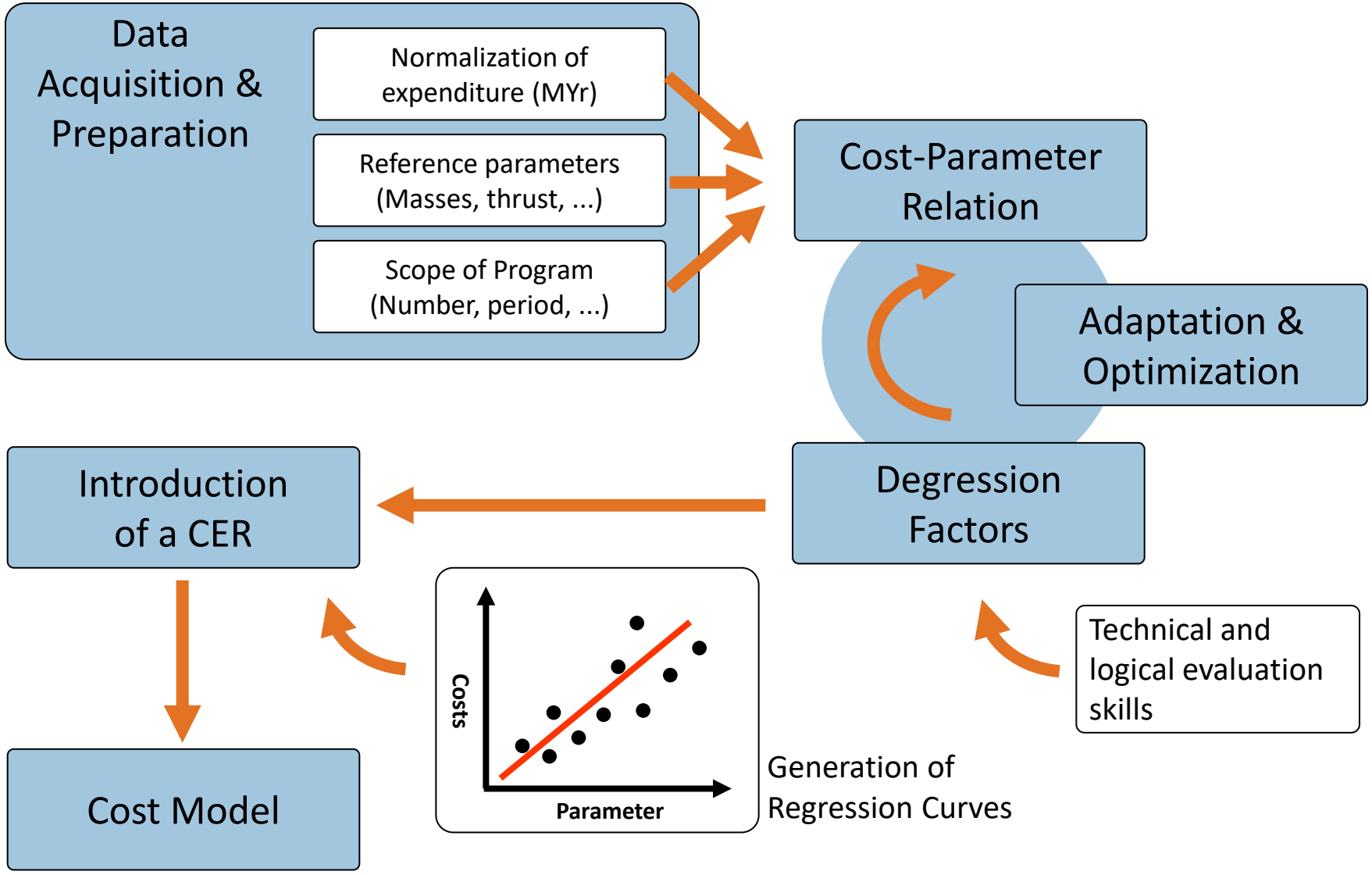


Cost Estimation in the Preliminary & Detail Design Stage

In the preliminary design stage, a holistic cost model is preferred that can be used for evaluation and decision purposes.



Derivation of the Parametric Cost Model



Cost-Estimation Relation (CER)

Definition: The Cost-Estimation Relation is the mathematical expression of the relationship between the costs (which is the dependent variable) and another independent, cost-driving parameter.

$$\Gamma = a \cdot P^x$$

Γ: Costs

a: System-specific constant

P: Independent cost-driving parameter (mass, thrust, etc.)

X: System-specific sensitivity factor

In aviation projects, the mass is frequently used as a cost-driving parameter.

Degression Factors

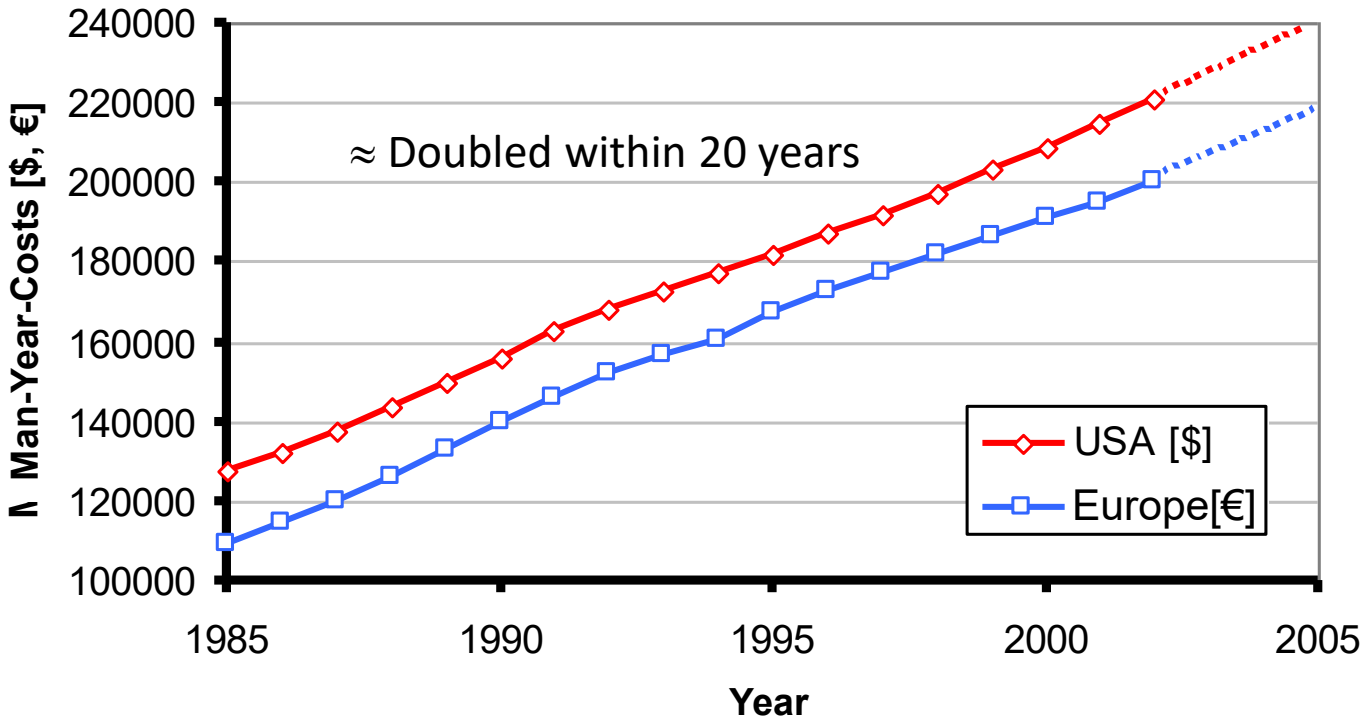
- When developing CERs, those factors have to be taken into account that may depend on technical or non-technical aspects and hence influence the costs strongly.
- That is why **degression factors** are introduced
  - $\mathbf{f_1}$:Factor of the technological state of the art (e.g., first system of its kind)
  - $\mathbf{f_2}$: Technical quality factor (describes system-inherent technical aspects)
  - $\mathbf{f_3}$: Experience-related factor (describes the experience of the teams with similar projects)
- Regarding the production costs (recurring costs), the learning factor has to be taken into account
  - $\mathbf{f_4}$: Cost-reduction factor due to series production

Using the Man-Year to define Costs

- The man-year enables cost estimations independent of
  - time-related influences
  - inflationary influences
  - different companies
  - different countries
  - different currencies
  - exchange rate fluctuations
- The conversion of man-year into a currency is done through a multiplication of the man-year costs with the reference value of the respective region and the respective year.
- The average value can be used as an approximation for purposes of development and production planning.

Man-Year Costs

Chronological development of the man-year-costs in the aerospace industry in Europe and the US.



Development Costs

The five different definitions of development costs

1. Unrealistic Development Costs

   Based on assumptions without cost estimations and experience, used to “sell” the concept.

2. Minimum Credible Development Costs

   Unrealistic cos estimations under competitive conditions, e.g. used to win a call for tenders.

3. Ideal or theoretical Development Costs (Bottom-Up)

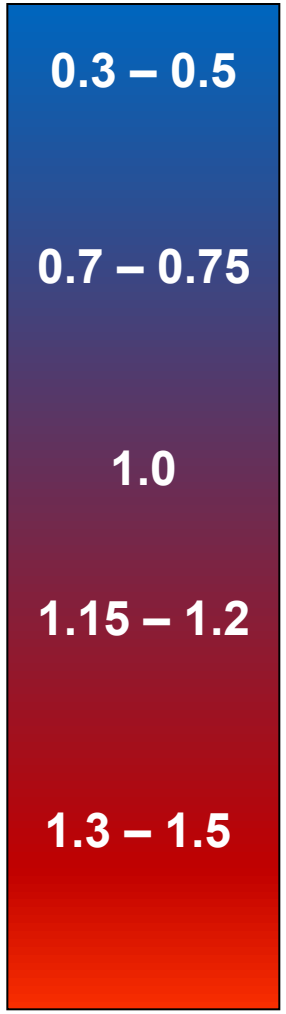
   Assuming that everything works as planned.

4. Realistic Development Costs

   Including the consideration of uncertainty with regard to unpredictable technical problems and delays.

5. Effective Costs until End of Development

   Overall costs after finishing the aircraft development program including inflationary influences.



Production Costs

The production phase is the step following the development phase. In principle, cost-planning functions are employed and determined in a way similar to planning the development costs.

- Assistance tasks and functions represent the main drivers for the production costs.
- Manufacturing costs (production of single components and parts, assembly…) represent the second cost driver.
- Cost of materials are only contributing by 5 – 10% to the production costs (even if expensive materials are employed).

